# REWARDING THE GRAPH BEHIND THE CHAIN: TOPOLOGY-AWARE PROCESS SUPERVISION FOR RL AND REASONING DISTILLATION

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## ABSTRACT

Long chain-of-thought reasoning is not purely linear: intermediate results are reused across non-adjacent steps, while unsupported conclusions may refer backward or form circular support. Outcome rewards and step-local process rewards do not expose these failures. We propose TopoPRM, a topology-aware implicit process reward model that first recovers an order-agnostic directed support graph from a free-form trace and then projects it onto a dependency DAG. Crucially, the backward and cyclic edges removed by this projection remain separate direction-consistency and acyclicity signals. We aggregate outcome, format, direction, acyclicity, and continuity with a topology-conditioned Choquet reward whose interactions explicitly encode structural agreement rather than averaging the two graph diagnostics into a generic topology score. The same projected graph guides distillation toward a compact dependency-preserving trace, without requiring graph-format generation from the policy. Matched counterfactuals verify that the signals are identifiable: reversing unchanged steps changes direction by $0.715$, matched cycle closure changes acyclicity by $0.328$ with zero direction shift, and premature averaging retains only $0.213\times$ the direction score's cross-trace variation. We pair these construct and edge-recovery audits with a fixed-scale, fixed-budget test of whether TopoPRM advances the accuracy–length Pareto frontier over outcome-only GRPO and additive or multiplicative aggregation.

## 1 INTRODUCTION

Reinforcement learning with verifiable rewards (RLVR) has substantially advanced long chain-of-thought (CoT) reasoning in large language models and now serves as a core mechanism for training mathematical and symbolic reasoning ability (Wei et al., 2022; Kojima et al., 2022; DeepSeek-AI, 2025; Yeo et al., 2025; Shao et al., 2024; Ke et al., 2026). From sparse final-answer rewards (Cobbe et al., 2021; DeepSeek-AI, 2025) to widely used process reward models (PRMs) that attach supervision to intermediate positions or prefixes (Lightman et al., 2023; Wang et al., 2024; Uesato et al., 2022), the dominant training interface still represents a trace as a *linear sequence of steps*. In practice, long reasoning is rarely linear: in mathematical derivations, a later conclusion often depends on several non-adjacent premises, an intermediate equation is reused by multiple downstream derivations, and subgoals branch before merging into the final answer (Besta et al., 2024; Zhong et al., 2026). Reconciling this structural mismatch has emerged as a core challenge for advancing reasoning supervision beyond linear credit assignment.

Recent reasoning models have made the macroscopic structure of inference clearer and more reliable, yet they remain confined to the CoT-as-result view. DeepSeek-R1 (DeepSeek-AI, 2025), DAPO (Yu et al., 2025), and Tulu 3 (Lambert et al., 2024) demonstrated that outcome-only rewards scale, but they could not distinguish a well-supported derivation from a trace that reached the same answer through unsupported intermediate claims, and they remained vulnerable to reward hacking (Gao et al., 2023; Skalse et al., 2022). On the PRM side, denser signals have been obtained through Monte Carlo rollouts (Wang et al., 2024; Luo et al., 2024), trained verifiers (Zhang et al., 2025; Khalifa et al., 2025; Zhao et al., 2026), and implicit outcome-derived scoring (Yuan et al., 2025; She et al., 2025). These methods can condition on a full prefix, but they do not explicitly recover the **multi-parent support relation** in which one intermediate result supports, or is supported by,

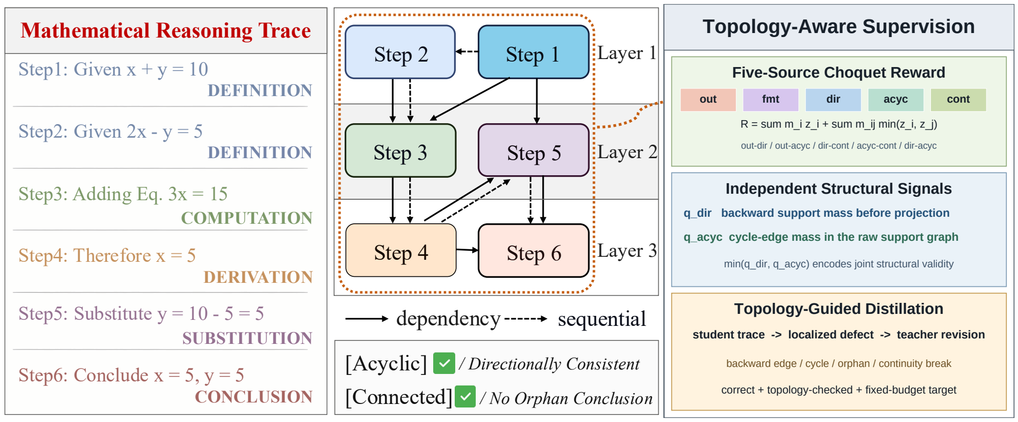


Figure 1: TopoPRM overview. A free-form mathematical trace is mapped to a layered dependency graph; direction and acyclicity enter the topology-conditioned multi-source reward, while the same graph diagnostics localize dependency-preserving teacher revisions for distillation. The policy itself never emits graph tags.

several non-adjacent steps (Wang et al., 2025b). Likewise, token-level distillation (Agarwal et al., 2024) does not require a compact target to preserve those cross-step dependencies.

We propose TopoPRM, which supervises reasoning through its *implicit topological dependency structure* rather than assuming that surface order is already valid (Figure 1). A frozen, order-agnostic edge encoder first recovers a raw directed support graph $\mathcal{H}_y$ from an unchanged free-form trace. Projecting $\mathcal{H}_y$ onto a text-order DAG $\mathcal{G}_y$ makes the discarded structure informative: backward-edge mass measures direction inconsistency, while cycle-edge mass measures circular support. We retain these as separate reward sources and aggregate them non-additively with outcome, format, and continuity. In particular, direction–acyclicity synergy rewards traces that satisfy both local support order and global non-circularity, while their outcome and continuity interactions make topology change rollout ordering rather than act as a small generic bonus. Finally, the same projected graph guides distillation toward a compact dependency-preserving trace. The policy never has to decode graph tags or a special DAG format. Our contributions are threefold:

1. **Raw support graph before acyclic projection.** We recover directed dependencies without imposing textual order, and turn the backward and cyclic structure removed by projection into explicit process supervision.
2. **Topology-conditioned multi-source optimization.** We introduce a monotone 2-additive Choquet reward that keeps direction and acyclicity independent and explicitly models their mutual, outcome, and continuity interactions, then reuse the graph diagnostics for Topology-Guided Distillation.
3. **Falsifiable structural and Pareto evaluation.** Matched reversal, cycle-closure, edge-noise, and premature-aggregation controls test whether direction and acyclicity are identifiable; an independent three-copy edge audit and fixed-scale, fixed-budget comparisons test recovery quality and the downstream accuracy–length frontier.

## 2 Related Work

**Reinforcement Learning with Verifiable Rewards.** RLVR replaces preference models with programmatic correctness checks on tasks whose answers are objectively verifiable (Shao et al., 2024; DeepSeek-AI, 2025; Lambert et al., 2024; Zeng et al., 2023). Recent work refines GRPO-style optimization through dynamic sampling and clipping (Yu et al., 2025), extended training schedules (Liu et al., 2025), finer credit assignment (Kazemnejad et al., 2025), asymmetric handling of negative samples (Zhu et al., 2025), entropy regularization (Cui et al., 2025), and simplified baselines (Ahmadian et al., 2024; Hu, 2025). Outcome-only signals nevertheless remain structurally blind: traces with very different dependency support can be reward-equivalent (Chen et al., 2026), and reward hacking persists even when the visible chain appears benign (Gao et al., 2023; Skalse et al., 2022; Wang et al., 2026b). *TopoPRM injects frozen topology and continuity diagnostics from latent reasoning graphs, targeting structural credit assignment directly rather than through a correctness surrogate.*

Concise-reasoning objectives introduce a separate confound: penalizing length inside a group-relative reward can assign negative advantage to a correct but long rollout. DRPO addresses this by decoupling the length-driven target distribution from correctness discrimination (Li et al., 2026a). DeCS localizes the necessary reasoning prefix with a separate detector, applies token-level reward only to the resulting chunks, and schedules prompt difficulty (Jiang et al., 2026); we reproduce its released 7B checkpoint under our fixed evaluator. LASER-D instead assigns step length rewards using targets that adapt to training-time behavior and per-question difficulty (Liu et al., 2026b); ARLCP coordinates difficulty-calibrated length and detected reflection penalties (Yu et al., 2026); and Self-Aligned Reward favors query-specific responses through a relative-perplexity signal rather than explicit trace structure (Han et al., 2026). Step-Tagging monitors semantic step types with lightweight classifiers and stops generation from their running counts (Belkhiter et al., 2025); it controls inference behavior but does not recover directed support among steps. These methods target redundancy or response efficiency, whereas our direction and acyclicity scores distinguish support relations that can occur at equal length and with unchanged wording. For multi-source reward integration, fair and stable reward composition alternates between mirror-descent weight updates and policy updates but retains a dynamic weighted sum (Li et al., 2024); HERO stratifies dense reward-model scores by binary verifier outcomes and reweights prompts by within-group variance (Tao et al., 2026). The former adapts source weights without representing source complementarity, while the latter preserves correctness ordering and makes dense feedback explicitly correctness-first. Our hypothesis is different: raw direction and acyclicity should remain identifiable, interacting reward sources rather than only additive terms or tie-breakers inside correctness strata. We therefore train a Shapley-matched additive control, compare additive, multiplicative, and Choquet rankings on one frozen candidate pool, and treat accuracy and length as co-primary policy outcomes.

**Process Reward Models for Reasoning.** PRMs score intermediate steps to address the sparsity of outcome rewards, originally relying on dense human annotation (Uesato et al., 2022; Lightman et al., 2023). Subsequent work reduces labeling cost via Monte Carlo or tree-search rollouts (Wang et al., 2024; Luo et al., 2024; Zhang et al., 2024; Lu et al., 2024; Zhang et al., 2025; Guan et al., 2025), generative PRMs make verifier reasoning explicit (Khalifa et al., 2025; Zhao et al., 2026; She et al., 2025), while recent process rewards use intrinsic semantic cues or explicitly tagged meta-reasoning actions (He et al., 2026; Zhang et al., 2026c). Process advantage verifiers tie step values to rollout-induced returns (Setlur et al., 2025; Lai et al., 2024); implicit PRMs derive them from outcomes alone (Yuan et al., 2025; Snell et al., 2024), and CRM conditions each step on the full preceding sequence while linking it to the final outcome (Zhang et al., 2026b). Recent benchmarks expose persistent weaknesses on fine-grained error categories and inter-step consistency (Song et al., 2025; Zheng et al., 2025). Such scorers can model a prefix but do not recover an explicit many-to-many support relation among steps. *TopoPRM requires no tagged policy format and trains no scalar neural process verifier; it derives frozen global-structure supervision from a recovered dependency graph together with local continuity.*

**Structured and Graph-Based Reasoning.** PARC links a completed chain to preceding premises for error identification (Mukherjee et al., 2025), while GRiD generates knowledge/reasoning nodes and verifies each node against declared parents (Wen et al., 2025). Graph of Verification adaptively decomposes a trace into a DAG and verifies node blocks in topological order without training (Fang et al., 2026). Other methods require graph-formatted reasoning: DAG-Math evaluates graph-level logical closeness (Zhang et al., 2026a), Structured Reasoning rewards within-graph flow and cross-sample overlap (Dong et al., 2026a), Structural CoT trains executable computation graphs (Anonymous, 2026b), VeriGraph makes a data-analytic agent emit a heterogeneous evidence DAG and rewards its integrity and coherence (Jin et al., 2026), the Graph Reasoning Paradigm emits labeled symbolic graphs (Liu et al., 2026a), and GoT-R1 reinforces parsability, sparsity, and non-linearity of serialized reasoning graphs with an additive reward (Li et al., 2026b). AtomGraph instead learns a graph-attention verifier over reasoning chains that have already been parsed into DAGs (Karmore, 2026). Contract-checked graph editing enforces acyclicity as a hard gate for composing candidate DAGs during inference-time search (Li & Cao, 2026). TUR-DPO sanitizes extracted claim-support graphs and linearly combines valid-path, cycle, dangling-node, and contradiction terms to shape offline preference optimization (Abdullah et al., 2026); it neither retains edge-direction residuals nor separates direction from cyclic support. RewardFlow merges observed agent trajectories into an environment-state transition graph and propagates terminal reward along action edges (Feng et al., 2026); it does not recover support direction within one free-text derivation. FlowTracer instead derives token credit from an attention-induced answer-flow DAG (Dong et al., 2026b). Two methods named Graph-GRPO propagate credit over a predefined e-commerce relevance graph or optimize sampled multi-agent communication topologies, rather than dependencies within one free-form trace (Che et al., 2026; Cang et al., 2026). TopoPRM instead leaves policy traces in free text and recovers an unconstrained semantic-support graph before projection. Declared-parent, attention-DAG, learned-DAG, valid-DAG, and hard-gated representations rule out or do not score backward and cyclic semantic support; we retain these residuals as direction and circular-justification signals.

Post-hoc graph probes support non-sequential reasoning (Zhong et al., 2026; Matsutani et al., 2026); CoT2Graph uses recovered graphs for failure analysis, outcome prediction, and Best-of-$N$ selection (Anonymous, 2026a), and Reasoning Consensus merges several recovered DAGs for inference-time structural ensembling (Parulekar et al., 2026). SARL clusters reasoning steps into latent functions and rewards clustering and short paths in their *undirected* transition graph, explicitly abstracting away directional control flow (Wang et al., 2026c). This small-world prior is complementary to our semantic-support relation: our direction score detects backward dependency mass, while acyclicity detects global self-support. Hidden-state cycles can reflect useful recurrent visits (Minegishi et al., 2025), while LoopBench studies self-reinforcing textual repetition during generation (Duan et al., 2026). Our cycles instead denote circular semantic support in the committed derivation: we do not penalize exploration, revision, or repeated wording, only a final support relation in which a claim depends on itself. The novelty is therefore not "reasoning as a graph," but raw direction/cycle diagnostics as online, auditable reward sources without graph-formatted generation; Appendix G makes this design boundary explicit against the closest alternatives.

**Distillation and Reasoning Compression.** Knowledge distillation transfers teacher capability (Hinton et al., 2015); student-generated variants reduce train–test mismatch (Gu et al., 2024; Agarwal et al., 2024; Ko et al., 2024), including ExOPD, BOND, KDRL, and black-box GAD (Yang et al., 2026; Sessa et al., 2024; Xu et al., 2025; Ye et al., 2025). Long-CoT compression uses pruning, length penalties, or budget control (Xia et al., 2025; Luo et al., 2025; Arora & Zanette, 2026; Aggarwal & Welleck, 2025; Muennighoff et al., 2025); graph-based pruning removes weak branches from projected DAGs (Yuan et al., 2026), while Reasoning Scaffolding distills semantic operations (Wen et al., 2026). TopoPRM instead uses raw direction/cycle violations to elicit and filter teacher revisions, preserving dependencies that token budgets or post-projection pruning cannot identify. *Topology-Guided Distillation is evaluated by the accuracy–length frontier and independently audited topology preservation.*

## 3 Method

TopoPRM (Figure 2) is a topology-aware post-training framework whose stages share a graph encoder $\mathcal{E}_\phi$ and an acyclic projection $\Pi$. Stage I produces parsable reasoning traces. Stage II applies GRPO to a topology-conditioned multi-source reward. Stage III performs Topology-Guided Distillation, in which the Stage-II teacher revises student rollouts using defects localized on the projected graph.

### 3.1 Problem Setup

Let $x = (q, c)$ pair a mathematical problem $q$ with optional context $c$, and let $\pi_\theta$ generate a trace $y = (s_1, \ldots, s_T)$. The frozen encoder produces a raw directed support graph $\mathcal{H}_y = \mathcal{E}_\phi(y) = (\mathcal{V}, \mathcal{A}_H)$, where $i \to j$ means that step $s_i$ supplies a result, claim, sub-goal, or assumption used by $s_j$. Unlike a DAG constructed only from pairs $i < j$, $\mathcal{H}_y$ may contain backward edges and cycles. We retain these violations and separately form $\mathcal{G}_y = \Pi(\mathcal{H}_y)$ by keeping support edges consistent with the trace order. Thus $\mathcal{G}_y$ supports DAG operations, while the projection cost of $\mathcal{H}_y$ exposes structural failures that would otherwise be removed by construction.

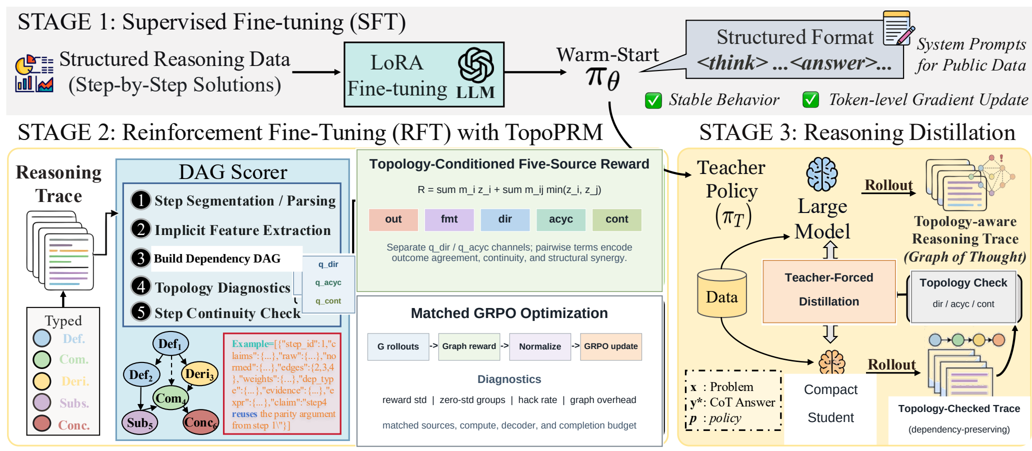


Figure 2: Overview of TopoPRM and Topology-Guided Distillation. An order-agnostic encoder recovers a raw directed support graph; its acyclic projection provides the training DAG while discarded backward and cyclic edges remain reward signals. A topology-conditioned Choquet reward aggregates multiple sources, and the projected graph guides dependency-preserving distillation.

**Scope of the acyclicity assumption.** Acyclicity is normative only for the *semantic support relation in the committed trace*. It does not forbid exploratory computation, checking an earlier result, or revising a failed branch. Such behavior may revisit a latent state or repeat a concept while still yielding an acyclic final justification. A cycle in $\mathcal{H}_y$ has the narrower meaning that a written claim ultimately supports itself; $q_{\text{acyc}}$ targets this circular justification rather than recurrent reasoning in general.

### 3.2 Topology-Aware Data Processing

**Raw-Graph Extraction and Projection.** Given a rollout $y$, $\mathcal{E}_\phi$ proceeds in three stages. **(i) Segmentation and parsing.** It splits $y$ into steps and sub-questions, then extracts normalized expressions, claim keys, variables, and step types. **(ii) Order-agnostic support encoding.** Auditable rules propose high-precision symbolic candidates, while a lightweight pair encoder predicts one of $\{i \to j, j \to i, \varnothing\}$ for each unordered intra-component pair. The encoder is trained offline from semantic-teacher edges and frozen during policy optimization; a separately annotated set calibrates edge precision and direction. Retained encoder edges carry their calibrated confidence, while rule edges use unit weight. Semantic predictions override an order-based rule when both judge the same pair, preventing artificial two-cycles caused by conflicting extractors. **(iii) Acyclic projection.** We preserve all predicted edges in $\mathcal{H}_y$, set $\mathcal{G}_y = \Pi(\mathcal{H}_y)$ by removing $i \to j$ with $i \geq j$, and add an adaptive weak sequential edge only when a node remains unsupported. Full rules and abstention thresholds are in Appendix C.

**Topology and Continuity Diagnostics.** Let $\mathcal{A}_H^{\text{dep}}$ denote the dependency edges of $\mathcal{H}_y$, let $w_{ij} \geq 0$ be the confidence weight of edge $(i,j)$, and set $W_H = \sum_{(i,j)\in\mathcal{A}_H^{\text{dep}}} w_{ij}$. Let $\mathcal{A}_{\text{cyc}}$ contain the dependency edges whose endpoints lie in the same cyclic strongly connected component (size greater than one, or a self-loop), and let $c_H$ be the fraction of nodes incident to at least one dependency edge. We first compute $d_H$, the forward-edge mass, and $u_H$, the cycle-edge mass, then shrink both diagnostics toward 0.5 when edge coverage is low:

$$q_{\text{dir}}(\mathcal{H}_y) = c_H d_H + (1 - c_H)/2, \quad d_H = \frac{\sum_{(i,j)\in\mathcal{A}_H^{\text{dep}}} w_{ij} \mathbb{1}[i < j]}{W_H}, \tag{1}$$

$$q_{\text{acyc}}(\mathcal{H}_y) = c_H(1 - u_H) + (1 - c_H)/2, \quad u_H = \frac{\sum_{(i,j)\in\mathcal{A}_{\text{cyc}}} w_{ij}}{W_H}. \tag{2}$$

When $W_H = 0$, we use $d_H = 1$ and $u_H = 0$; because $c_H = 0$ for a multi-step empty graph, both coverage-adjusted scores are nevertheless 0.5, not perfect. (A singleton trace has no possible cross-step violation and uses $c_H = 1$.) The continuity score $q_{\text{cont}}$ measures whether each step is traceable to the question or a supported prefix of $\mathcal{G}_y$. Direction and acyclicity remain independent properties of the raw graph rather than being averaged into one topology score or checked tautologically after projection. They are related but not redundant: every directed cycle contains a backward edge under the textual order, whereas a backward edge need not close a cycle, and raw graphs with identical backward-edge mass can have different cycle-edge mass. Thus $q_{\mathrm{dir}}$ detects local order inconsistency while $q_{\mathrm{acyc}}$ detects global circular support.

**Topology-Conditioned Multi-Source Aggregation.** Let $\mathbf{z} = (r_{\mathrm{out}}, r_{\mathrm{fmt}}, q_{\mathrm{dir}}, q_{\mathrm{acyc}}, q_{\mathrm{cont}}) \in [0,1]^5$. A weighted sum assumes that each source has the same marginal value regardless of the others. Instead, we use a monotone 2-additive Choquet model with Möbius coefficients (Choquet, 1954; Grabisch, 1997)

$$r_{\mathrm{total}}(x,y) = \sum_i m_i z_i + \sum_{(i,j)\in\mathcal{I}_{\mathrm{topo}}} m_{ij} \min(z_i, z_j), \quad \mathcal{I}_{\mathrm{topo}} = \{(o,d),(o,a),(d,c),(a,c),(d,a)\}. \tag{3}$$

Here $o, d, a, c$ denote outcome, direction, acyclicity, and continuity. All $m_i, m_{ij} \geq 0$ and sum to one, so the reward is bounded and monotone. The five interactions assign additional value when each raw-graph diagnostic agrees with outcome or continuity and when direction agrees with acyclicity. This makes topology a constituent of cross-source utility rather than a correctness-first bonus; no length source enters the main reward. We ablate the capacity against equal-weight, Shapley-matched additive, and multiplicative alternatives, and remove the direction–acyclicity interaction in isolation.

### 3.3 Topology-Guided Post-Training

**Stage I: Supervised Warm-Start.** We perform Supervised Fine-Tuning (SFT) on structured mathematical reasoning data to teach the policy to produce stepwise solutions parsable by $\mathcal{E}_\phi$. This reduces invalid rollouts during subsequent reinforcement learning, since topology diagnostics are meaningful only when the trace contains recoverable reasoning steps.

**Stage II: GRPO with TopoPRM Rewards.** Starting from the warm-start checkpoint, we optimise the policy with GRPO. For each completion, $\mathcal{E}_\phi$ constructs $\mathcal{H}_y$ and $\mathcal{G}_y$; Eq. 3 then scores the completion. Standard group normalization acts on the complete Choquet reward, so topology can change the ordering of rollouts rather than only break ties inside an outcome stratum. Reward-hacking tests separately measure whether structurally plausible but answer-incorrect traces are over-ranked.

**Stage III: Topology-Guided Distillation.** The optimized model after Stage II serves as the teacher $\pi_T$, while a smaller pretrained checkpoint initializes the student $\pi_S$. Topology-Guided Distillation conditions target construction on topology diagnostics from student-generated traces. For each prompt $x$, the student first rolls out $y_{\mathrm{init}} \sim \pi_S(\cdot \mid x)$; $\mathcal{E}_\phi$ produces $\mathcal{H}_{y_{\mathrm{init}}}$ and $\mathcal{G}_{y_{\mathrm{init}}}$; a dispatcher $\psi$ then issues a revision instruction targeting a backward edge, cyclic component, orphan conclusion, or continuity break. The teacher generates a revised target under this instruction. Thus topology determines where supervision is requested and which revisions are retained, rather than serving only as a post-hoc score.

**Distillation objective.** Let $A(y_{\mathrm{init}}, y_{\mathrm{rev}})$ indicate that the revised trace has a correct boxed answer and valid format, satisfies $q^{\mathrm{rev}}_{\mathrm{dir}} \geq \tau_d$, $q^{\mathrm{rev}}_{\mathrm{acyc}} \geq \tau_a$, and $q^{\mathrm{rev}}_{\mathrm{cont}} \geq \tau_c$, does not reduce either raw-graph diagnostic relative to the initial trace, and fits a fixed token budget $B$. The retained dataset is $\mathcal{D}_{\mathrm{distill}} = \{(x, y_{\mathrm{rev}}) : A(y_{\mathrm{init}}, y_{\mathrm{rev}}) = 1\}$. The student minimises standard teacher-forced distillation loss

$$\mathcal{L}_{\mathrm{distill}}(\theta) = -\mathbb{E}_{(x,y)\sim\mathcal{D}_{\mathrm{distill}}} \sum_{t=1}^{|y|} \log \pi_S(y_t \mid x, y_{<t}). \tag{4}$$

Unlike static trace distillation, the target distribution is concentrated on revisions elicited by the current student's structural failures. The explicit direction and acyclicity constraints prevent apparent compression from being obtained by deleting necessary support or introducing circular justification. The full pipeline is summarised in Appendix Algorithm 1.

## 4 EXPERIMENTS

**Draft audit status.** Red cells are not claims. Each method uses one canonical seed-0 training run; reported intervals quantify evaluation-set uncertainty, not training variance. The original EMNLP tables remain archival in Appendix H until they satisfy the canonical protocol below.

### 4.1 CANONICAL PROTOCOL

**Fixed model scale and budget.** The primary comparison starts from the same Qwen3.5-9B Stage-I checkpoint and changes only the Stage-II reward. The public frontier is restricted to dense released checkpoints with 7B–9B parameters; sparse models with similar total but much smaller active parameter counts are outside this matched scale. All variants use rank-64 LoRA, 300 GRPO steps, four generations per prompt, learning rate $5\times10^{-6}$, $\beta$=0.04, and a 4,096-token completion budget. Stage I uses 18,487 structured math solutions for training and 983 source-disjoint solutions for validation, three epochs, learning rate $5\times10^{-5}$, and global batch 32. Stage II uses 14,893 unique public train-split questions after exact normalized-question exclusion against every primary evaluation set. We remove 76 of 14,969 leakage-free candidates whose exact Qwen3.5 chat prompts exceed 512 tokens, rather than truncating their problem statements. Reference-solution DAGs are withheld because their step indices are not aligned with generated traces; $q_{\mathrm{dir}}$, $q_{\mathrm{acyc}}$, and $q_{\mathrm{cont}}$ are computed only from each rollout's $\mathcal{H}_y$ and $\mathcal{G}_y$. The pair encoder is frozen in every topology-aware run; the 9B semantic teacher is used only to create its offline supervision and never appears in the GRPO reward loop.

**Benchmarks and metrics.** The primary fixed-scale frontier uses GSM8K, MATH-500, the official 675-item English text-only mathematics subset of OlympiadBench, and AIME 2024, spanning elementary through competition mathematics. Pass@1 exact-answer accuracy and mean visible response tokens are co-primary; correctness is read only from the last non-empty box or a terminal explicitly marked answer, so intermediate expressions in truncated traces receive no credit. EOS padding is excluded by retokenizing each decoded response without special tokens. Method $A$ Pareto-dominates $B$ only when it is no less accurate and no longer, with one strict inequality; pass@5 and maj@5 are secondary diagnostics. Every reproduced row uses one seed-0 sampled response per item with temperature 0.6, top-$p$=0.95, top-$k$=20, min-$p$=0, repetition penalty 1.0, identical task content and verifier, and a 4,096-token cap. Released chat templates and documented role/thinking controls are retained; matched rows share tokenizer and message roles (Appendix A.4).

**Statistics and provenance.** Every row maps to a checkpoint, manifest, and per-example output. Marginal benchmark accuracy uses Wilson 95% intervals; token means and stratified macros use item bootstrap, while matched accuracy and length differences use paired bootstrap over shared items. With one training run, these quantify evaluation-set uncertainty only. Public checkpoints with longer budgets or repeated decoding are re-evaluated under our single-response 4,096-token protocol rather than quoted in Table 1. The human audit reports agreement, edge precision/recall, direction accuracy, and abstention coverage separately from teacher-held-out validation.

### 4.2 ACCURACY–LENGTH FRONTIER (RQ1)

Table 1 tests the central empirical claim: at a fixed 9B adaptation scale and response budget, does the raw-directed TopoPRM reward move the accuracy–length frontier beyond Stage-I SFT, outcome-only GRPO, the dynamic difficulty-aware LASER-D baseline, and a Shapley-matched additive multi-source reward? Our matched LASER-D adaptation preserves the released mechanism: training-time length coverage determines a target for each rollout-difficulty band, and only correct responses within that target receive its step reward (Liu et al., 2026b); Appendix B records the budget-scaled thresholds and monitor protocol. The completed Stage-I row has macro accuracy 42.1% (stratified item-bootstrap 95% interval $[40.2, 44.4]$) and 526 visible tokens ($[423, 647]$); these intervals quantify evaluation-set uncertainty, not training variance. A SOTA claim will be made only if TopoPRM is non-dominated against all matched rows and improves the reproduced accuracy–length frontier over the listed 7B–9B public checkpoints. Until then, the table remains a preregistered comparison rather than a result narrative. *External bars.* FlowTracer uses an attention-induced answer-flow DAG under a 1K-output protocol (Dong et al., 2026b); Structured Reasoning

Table 1: Fixed-budget accuracy–length comparison at dense 7B–9B scale. Cells report pass@1 (%) and mean visible response tokens from one seed-0 sample per item under matched sampling, task content, and a 4,096-token cap; released chat templates and required thinking controls are retained. Adapted methods use one canonical seed-0 training run. Accuracy intervals are Wilson; token and paired-difference intervals use item bootstrap. Neither measures training variance.

| | GSM8K | | MATH-500 | | OlympiadBench | | AIME 2024 | | Macro avg. | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | Acc. | Tok. | Acc. | Tok. | Acc. | Tok. | Acc. | Tok. | Acc. | Tok. |
| *Public checkpoints, reproduced with the same evaluator* | | | | | | | | | | |
| Qwen3-8B (Qwen Team, 2025) | 91.4 | 1,948 | 64.6 | 3,185 | – | – | 0.0 | 4,096 | – | – |
| SpyRL-Qwen3-8B-Math (Wang et al., 2026a) | – | – | – | – | – | – | – | – | – | – |
| GoT-R1-8B (Li et al., 2026b) | 95.2 | 357 | – | – | – | – | 16.7 | 2,018 | – | – |
| DeepSeek-R1-0528-Qwen3-8B (DeepSeek-AI, 2025) | – | – | – | – | – | – | – | – | – | – |
| DeCS-7B (Jiang et al., 2026) | – | – | – | – | – | – | – | – | – | – |
| MiMo-7B-RL-0530 (Xiaomi LLM Core Team, 2025) | – | – | – | – | – | – | – | – | – | – |
| Nemotron-Cascade-8B-Thinking (Wang et al., 2025a) | – | – | – | – | – | – | – | – | – | – |
| Klear-Reasoner-8B (Su et al., 2025) | – | – | – | – | – | – | – | – | – | – |
| OpenReasoning-Nemotron-7B (NVIDIA, 2025) | 83.0 | 2,429 | 51.2 | 3,260 | – | – | 0.0 | 4,096 | – | – |
| AceMath-RL-Nemotron-7B (NVIDIA, 2026) | – | – | – | – | – | – | – | – | – | – |
| Qwen3.5-9B (Qwen Team, 2026) | 76.4 | 2,710 | 40.4 | 3,627 | – | – | 0.0 | 4,096 | – | – |
| *Matched Qwen3.5-9B adaptation from the same Stage-I checkpoint* | | | | | | | | | | |
| Stage-I SFT | 85.8 | 137 | 59.6 | 283 | 19.6 | 604 | 3.3 | 1,080 | 42.1 | 526 |
| Outcome-only GRPO | – | – | – | – | – | – | – | – | – | – |
| LASER-D (matched) (Liu et al., 2026b) | – | – | – | – | – | – | – | – | – | – |
| Matched additive reward | – | – | – | – | – | – | – | – | – | – |
| TopoPRM (Choquet) | – | – | – | – | – | – | – | – | – | – |

Table 2: Direction and acyclicity mechanism tests. **Top:** matched policy ablations use one decoder and budget. **Bottom:** targeted frozen-encoder counterfactuals; Appendix Table 7 gives the complete ledger and paired intervals. Red cells await canonical policy outputs.

| | Accuracy (%) | | | | Aggregate | | Generated topology | |
|---|---|---|---|---|---|---|---|---|
| **Policy variant** | GSM8K | MATH | Olymp. | AIME | Macro | Tok. | $q_{\mathrm{dir}}$ | $q_{\mathrm{acyc}}$ |
| Full TopoPRM | – | – | – | – | – | – | – | – |
| w/o direction source | – | – | – | – | – | – | – | – |
| w/o acyclicity source | – | – | – | – | – | – | – | – |
| w/o raw support | – | – | – | – | – | – | – | – |

*Frozen-encoder counterfactuals on identical source traces*

| **Control** | $n$ | **Direction response** | **Acyclicity response** | **Isolated effect** |
|---|---|---|---|---|
| Reverse all steps | 120 | $\Delta q_{\mathrm{dir}} = -.715$ | $\Delta q_{\mathrm{acyc}} = +.064$ | Textual order / direction |
| Matched cycle closure | 57 | $\Delta q_{\mathrm{dir}} = .000$ | $\Delta q_{\mathrm{acyc}} = -.328$ | Cycle at fixed backward mass |
| Premature averaging | 120 | $\sigma_{\mathrm{dir}} = .110$ | $\sigma_{\mathrm{acyc}} = .107$ | Avg. $\sigma = .023$ ($.213\times$); opposite: 39.2% |

applies attention-MaxFlow or cross-response LCS rewards to explicitly tagged traces (Dong et al., 2026a); and the Graph Reasoning Paradigm makes the policy emit labeled graph nodes, reporting a 7,040-token AIME mean that exceeds our cap (Liu et al., 2026a). ARLCP and Self-Aligned Reward instead target efficient reasoning without explicit support direction (Yu et al., 2026; Han et al., 2026). Because their models, training, and decoders differ, these are external bars rather than rows mixed into the reproduced Pareto comparison.

### 4.3 Where the Gain Comes From (RQ2)

**Raw direction and acyclicity.** Table 2 pairs policy ablations with matched counterfactuals. Reversal changes $(q_{\mathrm{dir}}, q_{\mathrm{acyc}})$ by $(-0.715, +0.064)$. Their paired 95% intervals are $[0.657, 0.771]$ and $[0.040, 0.091]$, consistent with changed order but no added cycle. Across 57 edge-count, backward-mass, and coverage-matched pairs, cycle closure changes only $q_{\mathrm{acyc}}$ by $-0.328$ ($[0.292, 0.366]$). Direction-flip and false-edge controls selectively reduce $q_{\mathrm{dir}}$ by 0.180 and $q_{\mathrm{acyc}}$ by 0.190 (Appendix Table 7; Figure 3).

Table 3: Frozen four-candidate aggregation diagnostic. Rows rank identical Stage-I samples; benchmark cells are selection, not trained-policy, accuracy. AUC measures correct–incorrect pair ranking and StructWrong counts wrong selections with $q_{\mathrm{dir}}, q_{\mathrm{acyc}} \geq .8$. Cells await the canonical pool; intervals are item bootstrap, not training variance.

| **Aggregator** | **GSM8K** | **MATH** | **Olymp.** | **AIME** | **Macro** | **AUC**$\uparrow$ | **StructWrong**$\downarrow$ | **Tok.**$\downarrow$ | $q_{\mathrm{dir}}\uparrow$ | $q_{\mathrm{acyc}}\uparrow$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Outcome only | – | – | – | – | – | – | – | – | – | – |
| Equal-weight additive | – | – | – | – | – | – | – | – | – | – |
| Shapley-matched additive | – | – | – | – | – | – | – | – | – | – |
| Shapley-matched multiplicative | – | – | – | – | – | – | – | – | – | – |
| Choquet w/o dir.–acyc. term | – | – | – | – | – | – | – | – | – | – |
| Topology-conditional Choquet | – | – | – | – | – | – | – | – | – | – |

Table 4: Independent three-copy blind edge audit. Cells await annotation.

| | | **Overall edge recovery** | | | **Topology validity** | | | **Edge F1 by trace length** | | | **Agreement** |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Extractor** | $n$ | Prec. | Rec. | F1 | Dir. | Cyc. | Cov. | Short | Medium | Long | $\kappa$ |
| Adjacent-step chain | 120 | – | – | – | – | – | 1.00 | – | – | – | – |
| Rules only | 120 | – | – | – | – | – | – | – | – | – | – |
| Pair encoder (no abstention) | 120 | – | – | – | – | – | 1.00 | – | – | – | – |
| Frozen pair encoder (calibrated) | 120 | – | – | – | – | – | – | – | – | – | – |
| Semantic teacher (audit upper bound) | 120 | – | – | – | – | – | – | – | – | – | – |
| Projected DAG only | 120 | – | – | – | – | – | – | – | – | – | – |
| Hybrid raw-graph encoder | 120 | – | – | – | – | – | – | – | – | – | – |

On reversed controls the scores are anticorrelated ($r = -0.907$, $[-0.976, -0.779]$): averaging shrinks cross-trace standard deviation from $0.110$ to $0.023$, a ratio of $0.213$ ($[0.110, 0.320]$), and 39.2% of traces move oppositely. Keeping $q_{\mathrm{dir}}$ and $q_{\mathrm{acyc}}$ separate therefore preserves textual-order and global self-support information that premature averaging cancels; the policy panel tests its downstream value.

**Non-additive aggregation.** Table 3 isolates scalar aggregation by ranking the same four Stage-I candidates. It cannot populate trained-policy cells: the formal comparison trains only full and Shapley-matched additive rewards. We retain topology interactions only if frozen selection and the trained frontier both improve over matched additive weighting.

### 4.4 Independent Structural Audit and Cost (RQ3)

The 120 traces balance original, reversed, and interleaved order (40 each). Three blinded copies share each trace's condition but independently shuffle presentation, supporting pairwise agreement and a strict-majority reference without revealing model edges, scores, or IDs. The dashboard exposes every graph and score calculation (Appendix Figure 4); Table 4 audits edge recovery, direction, cycles, coverage, and incorrect-plausible traces. Cost separates offline labeling/adaptation from online latency, memory, graph pairs/s, and GRPO share under the logged $4\times$L20 allocation.

## 5 Conclusion

We presented TopoPRM, which converts backward and cyclic support into direction and acyclicity supervision before DAG projection and reuses both signals for reward aggregation and distillation without graph-format decoding. Matched counterfactuals show that the signals respond differently and that premature averaging suppresses cross-trace variation. They depend on edge recovery and complement, rather than replace, semantic verification.

**Scope and limitations.** We supervise a *committed* derivation, not its hidden search. A coherent trace may still be false, so topology complements correctness and inherits edge-encoder errors; teacher metrics cannot replace the human audit. Claims are restricted to public mathematics, dense 7B–9B checkpoints, and one budget; Choquet and distillation claims remain conditional on matched non-dominance.

## Ethics Statement

The canonical study uses public mathematics benchmarks and collects no demographic attributes. Human annotators label directed support relations in reasoning traces; their identities are not part of the released annotations. An archival appendix table retains a prior private teacher-grading benchmark for continuity with the earlier submission, but it does not support the ICLR claims. Before submission, that table requires documented authorization, privacy review, and release constraints; otherwise it will be removed from the submitted appendix.

## Reproducibility Statement

This draft is being audited for implementation-to-equation consistency and row-level result provenance. The submission version will point to the exact method definitions, configurations, raw-result manifests, and anonymized code bundle used to reproduce every retained result.

## AI Use Statement

Generative AI tools assisted with prior-review analysis, literature search and synthesis, hypothesis and experimental-control refinement, method implementation and code review, dataset cleaning, reproducibility auditing, scientific-figure editing, and manuscript drafting and revision. Before submission, the authors will manually inspect the cited literature, review code and generated artifacts, and verify every reported result against raw outputs. The authors take responsibility for the final claims, code, artifacts, and results; this disclosure will be updated if additional uses occur before submission.

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# A Implementation Details

## A.1 Hyperparameters

Table 5 lists the key hyperparameters for each training stage.

Table 5: Hyperparameter summary for the three training stages.

| **Parameter** | **Value** | **Description** |
|---|---|---|
| *Stage I: Supervised Fine-Tuning (SFT)* | | |
| Base model | Qwen3.5-9B | – |
| Adapter | LoRA r=64, $\alpha$=128 | all-linear modules |
| Learning rate | $5 \times 10^{-5}$ | cosine, 3% warmup |
| Epochs | 3 | – |
| Selected checkpoint | step 1,100 | minimum held-out loss 0.381609; final step 1,734 |
| Batch size | 1/device, accum 8 | global batch 32 on 4 GPUs |
| Max seq length | 4096 | – |
| Precision | bf16 | – |
| *Stage II: GRPO with TopoPRM Rewards* | | |
| Base model | SFT step 1,100 | held-out-loss selection |
| Adapter | LoRA r=64, $\alpha$=128 | all-linear modules |
| Learning rate | $5 \times 10^{-6}$ | cosine, 5% warmup |
| Gradient steps | 300 | – |
| Batch size | 1/device, accum 4 | global prompt batch 16 on 4 GPUs |
| Num generations | 4 per prompt | for GRPO grouping |
| Temperature | 0.9 | rollout sampling |
| $\beta$ (KL penalty) | 0.04 | – |
| Max completion length | 4096 | prompt capped at 512 |
| Precision | bf16 | – |
| *Stage III: Topology-Guided Distillation* | | |
| Teacher model | Stage-II 9B | frozen during distillation |
| Student model | Qwen3.5-4B | canonical cross-scale run |
| Adapter | LoRA r=64, $\alpha$=128 | all-linear modules |
| Learning rate | $2 \times 10^{-5}$ | cosine, 3% warmup |
| Epochs | 1 | – |
| Batch size | 2/device, accum 4 | global batch 32 on 4 GPUs |
| Student temperature | 0.7 | student-sampled trace |
| Teacher temperature | 0.0 | greedy revision target |
| Target budget | 1024 tokens | fixed across methods |
| Target selection | answer, format, direction, acyclicity | threshold calibrated on validation |
| Max seq length | 4096 | – |
| Precision | bf16 | – |

### A.2 Reward Configuration

The Choquet reward in Eq. 3 uses the five independent sources $(r_{\text{out}}, r_{\text{fmt}}, q_{\text{dir}}, q_{\text{acyc}}, q_{\text{cont}})$ and the interaction set $\{(o,d),(o,a),(d,c),(a,c),(d,a)\}$. The exact non-negative capacity coefficients are pending author confirmation and are not yet a training or paper claim. Once locked, the coefficients will sum to one and remain fixed across the canonical run and its matched controls. The additive control uses the capacity's Shapley source importance; the multiplicative control uses those same exponents. No length or reference-solution graph enters the canonical main reward because generated traces have no node alignment with a reference solution.

### A.3 Hardware and Infrastructure

Canonical ICLR runs use 4×NVIDIA L20 GPUs. We report measured wall-clock time and peak memory from this configuration rather than extrapolating from the earlier A100 runs.

## A.4 Evaluation Protocol

All public benchmark evaluations use the Transformers generation backend. Pass@1 is one sampled response per item from a single seed-0 evaluation pass; this is neither a multi-seed training result nor an average over repeated decodes. The shared decoder uses temperature 0.6, top-$p$=0.95, top-$k$=20, min-$p$=0, and repetition penalty 1.0. This matches Qwen3 and DeepSeek-R1's reasoning-time recommendations and keeps the public and adapted rows comparable; it differs from Qwen3.5's general-task recommendation only in using temperature 0.6 instead of 1.0. Generation explicitly enables the standard key–value cache even if a released checkpoint retains a training-time `use_cache=false` setting; this is recorded in the result manifest. The four primary mathematics benchmarks use the same 4,096-token completion cap and the same instruction to provide a step-by-step solution with the final answer in `\boxed{}`. We retain each released chat template and documented control: GoT-R1, OpenReasoning-Nemotron, and AceMath receive a user-only task; MiMo uses an explicit empty system turn; the released Nemotron-Cascade template injects the terminal "`/think`" control; and the AceMath, Qwen-family, and Klear templates activate their released reasoning format without an evaluator-added prefix. No model receives an uncounted response prefix. For mathematics tasks, the evaluator verifies the last nonempty boxed answer, or a short terminal region after the last explicit final-answer marker, against the dataset target; it never searches an unsubmitted truncated derivation for a matching intermediate expression. The reference is passed first to the asymmetric `math_verify` verifier. Visible length is obtained by retokenizing the decoded response without special tokens. Pass@5 and maj@5, when run as secondary diagnostics, draw five samples from the same decoder and reuse this equivalence relation rather than raw string equality. The manifest stores the message-role protocol, prefix flag, decoder parameters, evaluation seed, batch size, model and adapter paths, dataset fingerprint, and full per-example responses; cached results are reused only when these protocol fields match.

# B Choquet Capacity and Baselines

For non-negative Möbius coefficients summing to one, Eq. 3 is a normalized monotone 2-additive capacity (Choquet, 1954; Grabisch, 1997): it maps $\mathbf{0}$ to 0 and $\mathbf{1}$ to 1 and remains in $[0, 1]$. We separately replace the direction–acyclicity interaction with equal singleton mass to preserve both normalization and Shapley source importance, distinguishing the proposed structural synergy from merely increasing either source's weight. We also test traces that are structurally plausible but answer-incorrect to detect reward hacking.

**Matched baseline protocols.** The LASER-D adaptation uses the true response-token count and four rollouts per prompt. It labels a prompt high, medium, or low by rollout accuracy thresholds $2/3$ and $1/3$, and adds 0.5 only to correct rollouts at or below the band's target. Every 20 steps, the target is recomputed from a synchronized rolling monitor of 500 prompts. To match our 4,096-token cap, the released 4,096–16,384 candidate range is scaled by four to 1,024–4,096 in increments of 256; no character-length proxy is used. The matched additive policy control uses the Choquet capacity's Shapley source importance, preserving marginal source weights while removing every interaction. Equal additive, matched multiplicative, and direction–acyclicity-interaction controls are evaluated on the identical frozen four-candidate pool rather than presented as separately trained policies.

# C DAG Extraction Details

**Extraction pipeline.** Given a reasoning trace in `<think>`, we apply five frozen stages:

1. Step segmentation: split by numbered-step markers, mathematical discourse markers ($\because$, $\therefore$), and line boundaries.
2. Sub-question partitioning: markers such as "Sub-question $k$" define connected components; cross-sub-question edges are disallowed.
3. Information extraction: for each step, extract expressions, claims, and variables with rule-based patterns.
4. Raw support encoding: combine high-precision symbolic candidates with a lightweight pair classifier that selects $i \to j$, $j \to i$, or abstention without imposing text order.

Table 6: Detailed diagnostics for the frozen candidate pool in Table 3. Tie accuracy averages correctness across all top-scoring candidates rather than choosing the first; tie and zero-spread rates expose uninformative ranking groups. Cells await the canonical pool and report item-bootstrap intervals, not training variance.

| Aggregator | Tie acc.↑ | Oracle acc.↑ | Tie groups↓ | Zero-spread↓ | StructWrong↓ |
|---|---|---|---|---|---|
| Outcome only | – | – | – | – | – |
| Equal-weight additive | – | – | – | – | – |
| Shapley-matched additive | – | – | – | – | – |
| Shapley-matched multiplicative | – | – | – | – | – |
| Choquet w/o dir.–acyc. interaction | – | – | – | – | – |
| Topology-conditioned Choquet | – | – | – | – | – |

5. Projection: retain the raw graph for diagnostics, remove backward edges to obtain the training DAG, then add adaptive weak sequential edges only when support is absent.

**Information-density filters.** To avoid low-information nodes and noisy links: (a) claims are extracted at sentence level, with heading fragments and unfinished prefixes removed; (b) trivial tokens (single symbols, pure numerals, non-relational fragments) are excluded from dependency matching.

**Dependency rules.** For each unordered pair of steps in the same sub-question, rules propose candidates from informative expression reuse, canonicalized claim overlap, variable roles, and explicit step references. The pair classifier assigns direction or abstains. Only after raw-graph diagnostics are computed do we insert a fallback edge for an unsupported derivation or conclusion in the projected DAG.

**Pair-encoder adaptation.** The lightweight encoder is Qwen3.5-0.8B with a rank-8 LoRA three-way classification head. The frozen 9B semantic teacher processes 595 candidate source traces; 536 contain parsed semantic edges, and the shared strict segmentation filter rejects one dangling clause. The remaining 535 sources are hash-split before order augmentation into 428 training and 107 validation sources. After original, reversed, and interleaved presentations, deterministic negative subsampling yields 9,301 training pairs; the unsampled validation set contains 2,897 pairs. On these teacher-held-out pairs, the selected epoch-3 encoder reaches 0.7784 three-class macro-F1, 0.7711 edge precision, 0.7603 edge recall, and 0.7375 direction accuracy on gold edges. The calibrated edge-confidence and direction-margin thresholds are 0.70 and 0.20. This teacher-held-out diagnostic is not the independent human result in Table 4. The encoder is frozen during policy optimization; a retained encoder edge keeps its calibrated confidence as $w_{ij}$, while auditable rule edges use $w_{ij} = 1$. The semantic-teacher cache, adapter weights, corrected 120-trace audit pack, and all-order frozen prediction ledger have SHA-256 fingerprints `69131f16...b19c`, `9e8e78ac...3212`, `95c440a3...b53`, and `8c7a644e...9328`, respectively; all order-control statistics are paired by source record.

**Controlled edge noise.** Table 7 measures how the two raw-graph diagnostics respond when frozen predictions are corrupted after extraction. Deletion probes missed edges, false-edge addition probes spurious support, and direction reversal probes orientation errors. The audit uses one deterministic perturbation draw per source trace and paired bootstrap across traces; it is not a repeated-training or multi-seed result.

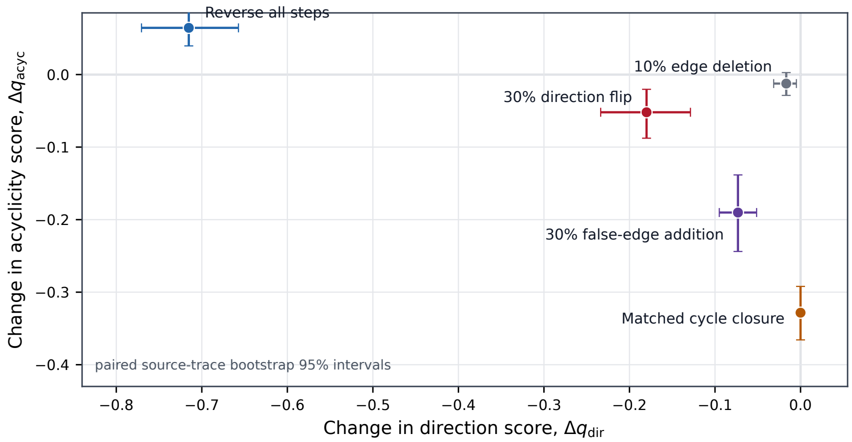


Figure 3: Controlled response of the frozen raw-graph encoder. Each point is the paired mean change relative to the identical source trace; bars are paired source-trace bootstrap 95% intervals. Reversal primarily changes direction, matched cycle closure changes acyclicity at fixed backward mass, and stochastic edge corruption moves both diagnostics according to its targeted failure. These are deterministic perturbation audits, not downstream policy results or training-seed variation.

Table 8: Diagnostic Best-of-8 selection accuracy (%) on one shared Qwen2.5-Math-7B-Instruct candidate pool ($n$=80 per benchmark; 2,048-token generation cap). PRM uses Qwen2.5-Math-PRM-7B; Hybrid multiplies normalized PRM by $1 + q_{\mathrm{topo}}$. Oracle reports whether any candidate is correct. This diagnostic is separate from the primary 4,096-token policy comparison.

| **Benchmark** | **Pass@1** | **Maj@8** | **PRM@8** | **Topo@8** | **Hybrid@8** | **Oracle@8** |
|---|---|---|---|---|---|---|
| GSM8K | 93.75 | 95.00 | 95.00 | 91.25 | 95.00 | 96.25 |
| MATH-500 | 83.75 | 85.00 | 83.75 | 78.75 | 83.75 | 92.50 |

Table 7: Frozen-encoder sensitivity on 120 source traces. One deterministic seed-0 perturbation is applied per trace; intervals are paired source-trace bootstrap 95% intervals. $p_{\mathrm{real}}$ is the realized modified-edge fraction. Deltas are corrupted minus original.

| **Perturbation** | $p$ | $p_{\mathrm{real}}$ | $\Delta q_{\mathrm{dir}}$ | $\Delta q_{\mathrm{acyc}}$ | $\Delta$Cov. | $\Delta$Cycle mass |
|---|---|---|---|---|---|---|
| Random edge deletion | .10 | .068 | $-.017$ $[-.031, -.005]$ | $-.012$ $[-.029, +.003]$ | $-.033$ | $-.006$ |
| | .20 | .161 | $-.027$ $[-.047, -.008]$ | $-.042$ $[-.061, -.023]$ | $-.091$ | $-.006$ |
| | .30 | .274 | $-.067$ $[-.092, -.044]$ | $-.073$ $[-.097, -.049]$ | $-.154$ | $-.005$ |
| Random false-edge addition | .10 | .046 | $-.013$ $[-.021, -.004]$ | $-.043$ $[-.065, -.023]$ | $+.008$ | $+.048$ |
| | .20 | .124 | $-.034$ $[-.050, -.019]$ | $-.099$ $[-.136, -.064]$ | $+.013$ | $+.111$ |
| | .30 | .244 | $-.073$ $[-.095, -.051]$ | $-.190$ $[-.244, -.138]$ | $+.031$ | $+.217$ |
| Random direction flip | .10 | .097 | $-.052$ $[-.084, -.023]$ | $-.018$ $[-.036, -.003]$ | .000 | $+.021$ |
| | .20 | .214 | $-.104$ $[-.153, -.057]$ | $-.034$ $[-.064, -.006]$ | .000 | $+.037$ |
| | .30 | .300 | $-.180$ $[-.234, -.129]$ | $-.052$ $[-.088, -.020]$ | .000 | $+.059$ |

**Topology is not a standalone correctness verifier.** Table 8 rescales the rebuttal-era shared Best-of-8 pool with the canonical gold-first symbolic verifier. Topology-only selection underperforms the dedicated PRM on both subsets, and their simple product ties rather than improves on the PRM. We therefore use topology as an auditable training-time structural source and do not claim that it replaces semantic verification. The candidate-pool and cached-score ledgers have SHA-256 fingerprints `2f4ded08...2664` and `8304d8fc...b7c8`.

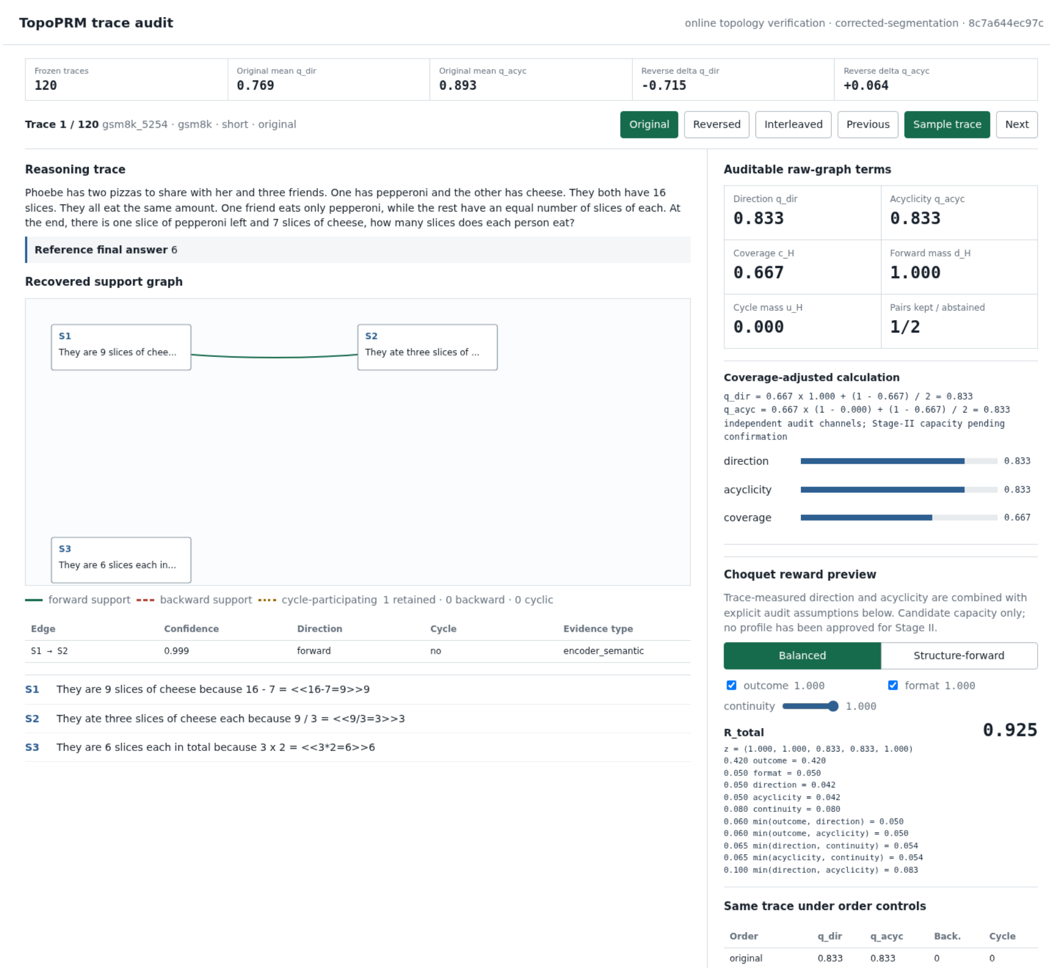


Figure 4: Trace-level topology audit interface. For each sampled reasoning trace, the dashboard exposes the recovered support graph, its edge ledger, and independently recomputed coverage-adjusted direction and acyclicity channels. The interactive reward preview substitutes these trace-measured channels into either preregistered candidate capacity while outcome, format, and continuity remain explicit user-controlled assumptions; it is not a canonical Stage-II result. Original, reversed, and interleaved controls switch the same trace online. Three linked annotation copies hide model predictions and internal record identities, independently shuffle trace presentation, and retain a matched step-order condition for agreement. The illustrated corrected-encoder trace is one auditable sample, not an aggregate evaluation result.

**Adaptive sequential weak edges.** A weak sequential edge (weight 0.3) is added only when the target step lacks strong incoming support and the source step does not already emit strong outgoing dependencies. This prevents chain-dominant artifacts in long traces.

**Node typing.** Each node is classified into DEFINITION, DERIVATION, COMPUTATION, CONCLUSION, AUXILIARY, SUBSTITUTION, or CASE_ANALYSIS by keyword rules.

**Relationship between DAG quality and reasoning compression.** We test whether traces with stronger independent graph diagnostics (acyclicity, low orphan ratio, and directional consistency) admit shorter compressed forms under Topology-Guided Distillation. Intuitively, a well-structured DAG has fewer redundant branches and shorter critical paths, so the transitive reduction yields a more compact chain. The raw-graph scores $q_{\mathrm{dir}}$ and $q_{\mathrm{acyc}}$ enter Eq. 3 independently, while the projected DAG supplies localized revision targets. This relationship is evaluated rather than assumed.

## D REWARD FUNCTION DETAILS

**Outcome reward.** Extracts the final mathematical answer from the last `\boxed{}` expression, falling back to an explicit final numeric expression when needed. Symbolic equivalence to the reference answer is checked with `math_verify`; a correct answer receives 1 and an incorrect or unparseable answer receives 0.

**Format reward.** The released Qwen3.5 template owns the opening <think> prefix, so it is part of the prompt rather than the scored continuation. The canonical reward therefore requires the continuation to contain non-empty reasoning, exactly one closing </think> tag, and a subsequent boxed answer; it assigns 1 only when all three conditions hold and 0 otherwise. A generated opening tag is rejected, preventing the policy from receiving credit for duplicating a prompt-owned control token. This `prefilled_think` protocol is fixed in the launch environment and recorded in the immutable run manifest.

**Length-only control.** The explicit-efficiency baseline uses linear decay from 1.0 (at $\leq$ 2,000 characters) to 0.0 (at $\geq$ 8,000 characters). This signal is absent from the main TopoPRM reward; all evaluation tables independently report tokenizer-token counts.

**Raw-graph rewards.** The GRPO reward consumes the coverage-adjusted direction and acyclicity scores from Eqs. (1)–(2) as two aligned sources from one frozen encoder pass. It never replaces them with a scalar average. Orphan conclusions on the projected DAG remain an interpretable defect record for distillation but are not a sixth source in the canonical Stage-II capacity.

**Continuity reward.** Uses regex-based expression matching to verify that each step references expressions or claims from prior steps or the problem statement. The continuity score $q_{\mathrm{cont}}$ (Eq. 5) measures the fraction of supported steps.

## E SUPPLEMENTARY MATHEMATICAL DEFINITIONS

**Independent topology sources.** Let $c_H$ be the fraction of nodes incident to a dependency edge in the raw graph, and let $d_H$ and $u_H$ be the confidence-weighted forward-edge and cycle-edge mass defined in Eqs. (1)–(2). The affine shrinkage by $c_H$ maps missing edge evidence to uncertainty at 0.5 rather than structural perfection. The two scores stay separate through aggregation: $q_{\mathrm{dir}}$ responds to local reversal without requiring a cycle, whereas $q_{\mathrm{acyc}}$ responds to strongly connected support even when backward mass is held fixed. Their interaction in Eq. 3 can reward joint satisfaction without cancelling their distinct counterfactual responses.

**Continuity score (full form).** The local continuity quality is

$$q_{\mathrm{cont}}(y) = \begin{cases} 1.0, & \eta(y) = 1, \\ 0.8\,\eta(y), & \eta(y) < 1, \end{cases} \qquad \eta(y) = \frac{1}{T}\sum_{i=1}^{T} \mathbb{1}[\mathrm{supported}(s_i, s_{<i}, q)]. \tag{5}$$

*Remark.* Eq. (5) is the engineering-tuned continuity term used in the implementation. It is a separate source in the Choquet reward rather than being folded into topology before aggregation.

**Topology-conditioned revision dispatch.** The revision instruction is generated by

$$P_r = \psi(r_{\mathrm{out}}, q_{\mathrm{dir}}, q_{\mathrm{acyc}}, \rho_{\mathrm{orphan}}, q_{\mathrm{cont}}, \ell), \tag{6}$$

where $\ell$ identifies the offending edge, cyclic component, or unsupported step when one can be localized.

## F TOPOLOGY-GUIDED DISTILLATION DETAILS

Topology-Guided Distillation is an optional compression stage whose goal is to retain the teacher's structural reasoning profile while reducing generation length in compact students. Table 9 compares

Table 9: Matched 9B-to-4B distillation protocol. All student methods use the same teacher, candidate prompts, accepted-target count, token budget, and optimizer budget. Accuracy is pass@1; Dir. and Acyc. are paired with the independent edge audit. Cells remain pending until the single canonical run; marginal accuracy uses Wilson intervals and matched differences use paired item-level bootstrap intervals, neither training variance.

| Method | GSM8K | MATH-500 | Avg. Tok | Dir. | Acyc. | Accept% |
|---|---|---|---|---|---|---|
| 9B teacher | – | – | – | – | – | – |
| 4B initialization | – | – | – | – | – | – |
| Static teacher traces | – | – | – | – | – | – |
| Generic revision | – | – | – | – | – | – |
| Length-only revision | – | – | – | – | – | – |
| Topology-Guided Distillation | – | – | – | – | – | – |

**Algorithm 1** Topology-Guided Distillation Training Pipeline

**Require:** Base policy $\pi_0$; data $\mathcal{D}_{\text{sft}}, \mathcal{D}_{\text{rl}}$; Choquet coefficients $\{m_i, m_{ij}\}$
**Ensure:** Teacher $\pi_T$, compact student $\pi_S$

$\pi_\theta \leftarrow \text{SFT}(\pi_0, \mathcal{D}_{\text{sft}})$ ▷ Stage I
**for** each GRPO step on $\mathcal{D}_{\text{rl}}$ **do** ▷ Stage II
  Sample $\{y^{(i)}\}_{i=1}^{G} \sim \pi_\theta(\cdot \mid x)$; build $\mathcal{H}^{(i)} = \mathcal{E}_\phi(y^{(i)})$ and $\mathcal{G}^{(i)} = \Pi(\mathcal{H}^{(i)})$
  Compute $r_{\text{out}}^{(i)}, r_{\text{fmt}}^{(i)}, q_{\text{dir}}^{(i)}, q_{\text{acyc}}^{(i)}, q_{\text{cont}}^{(i)}$
  Compute $r_{\text{total}}^{(i)}$ by Eq. 3; normalize within the GRPO group
  Update $\pi'_\theta$ with the GRPO objective
**end for**
$\pi_T \leftarrow \pi'_\theta$; init. $\pi_S$ from a smaller pretrained checkpoint
**for** each prompt $x \in \mathcal{D}_{\text{rl}}$ **do** ▷ Stage III
  $y_{\text{init}} \sim \pi_S(\cdot \mid x)$; extract defect record $F(y_{\text{init}})$ from $\mathcal{H}$ and $\mathcal{G}$
  $P_r \leftarrow \psi(F(y_{\text{init}}))$; sample $y_{\text{rev}} \sim \pi_T(\cdot \mid x, y_{\text{init}}, P_r)$
  Retain $(x, y_{\text{rev}})$ only if answer, format, direction, acyclicity, and budget constraints pass
  Update $\pi_S$ with $\mathcal{L}_{\text{distill}}$ (Eq. 4)
**end for**
**return** $(\pi_T, \pi_S)$

static traces, generic revision, length-only revision, and Topology-Guided Distillation under the same teacher, student initialization, accepted-target count, and budgets. Direction and acyclicity retention must survive the independent audit; this extension returns to the main paper only if the single canonical run supports a central claim.

**Why static trace distillation is insufficient.** Conventional trace distillation (Hinton et al., 2015; Gu et al., 2024) presents every student with the same teacher solutions. It therefore cannot target a student's particular missing dependency, and a length-only target can become shorter by deleting support that is necessary for the conclusion. Topology-Guided Distillation instead constructs supervision after observing the compact student's own trace.

**Defect localization and revision.** For each initial student attempt, we retain the raw edge list and projected DAG. The defect record contains the endpoints of backward edges, nodes in a non-trivial strongly connected component, orphan conclusions in the projected DAG, and unsupported local transitions. The dispatcher selects the highest-priority localized defect and asks the frozen Stage-II teacher to rewrite the solution within a fixed response budget while preserving a boxed final answer.

**Target acceptance.** A revised trace is retained only if it has a correct boxed answer and valid format, reaches the calibrated topology threshold, does not reduce coverage-adjusted direction consistency or acyclicity relative to the initial trace, and stays within the shared token budget. We log every rejection reason and report acceptance coverage. This makes direction and acyclicity part of the distillation data contract, not merely an evaluation diagnostic.

**Student training and controls.** The 4B student is trained on accepted revisions with Eq. 4. Matched controls use the same teacher, prompts, number of accepted targets, and optimization budget but replace topology-conditioned dispatch with either a generic revision instruction or an explicit length-only instruction. External answer accuracy and generated tokens are primary; extractor-based structural retention is reported as internal-consistency evidence and paired with the independent human edge audit.

## G CLOSEST-METHOD DESIGN AUDIT

Table 10 isolates representation and optimization choices rather than comparing numbers obtained under incompatible models or evaluators. A method may use a graph without exposing violations that a DAG representation has already ruled out. Our distinguishing object is the unconstrained directed support graph before projection: backward and cycle-participating support remain separately measurable and enter the online reward through explicit source interactions.

Table 10: Design audit of the closest topology-based reasoning objectives. "Graph-form output" means that the policy must serialize a graph. "Raw dir." asks whether support edges that oppose text order remain measurable before a DAG constraint; "cycle signal" asks whether circular semantic support is separately scored. This is a mechanism comparison, not a performance claim.

| Method | Relation represented | Graph-form output | Raw dir. | Cycle signal | Online RL | Source interaction |
|---|---|---|---|---|---|---|
| GoT-R1 (Li et al., 2026b) | Declared predecessor in serialized graph | Yes | No | No | Yes | Additive |
| SARL (Wang et al., 2026c) | Undirected latent-function transition | No | No | No | Yes | Additive |
| TUR-DPO (Abdullah et al., 2026) | Elicited claim-support graph | No | No | Yes | No (DPO) | Additive |
| FlowTracer (Dong et al., 2026b) | Attention-induced answer-flow DAG | No | No | No | Yes | No |
| TopoPRM | Unconstrained directed semantic support | No | **Yes** | **Yes** | **Yes** | **2-additive** |

## H Retained EMNLP Submission Results

**Archival evidence, not current ICLR claims.** The nine result tables rendered in the EMNLP submission PDF are retained below with their numeric values unchanged. Retired method names are expanded or replaced only to avoid reintroducing abandoned terminology. These tables mix earlier checkpoints, decoders, response budgets, and in some cases explicitly marked approximate entries. The ICLR main tables use only the canonical single-run protocol in Section 4.1; archival rows will be promoted only after row-level provenance and protocol revalidation.

Table 11: Archived EMNLP Table 1: results on nine public benchmarks across four task categories (Elementary: GSM8K, MATH-500; Olympiad: OlympiadBench, Omni-Math; Competition: AIME'24, AIME'25, CNMO'24; General: MMLU, GPQA-Diamond). All numbers are the originally submitted pass@5 accuracies (%).

| | | Elementary | | Olympiad | | Competition | | | General | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Model** | **Params** | GSM8K | MATH | Olympiad | Omni | AIME'24 | AIME'25 | CNMO'24 | MMLU | GPQA-D | **Avg.** |
| *Reference open-source models* | | | | | | | | | | | |
| Qwen3.5-2B | 2B | 77.2 | 60.8 | 48.5 | 13.3 | 6.7 | 13.3 | 36.1 | 32.8 | 37.9 | 36.3 |
| Qwen3.5-4B | 4B | 85.1 | 55.8 | 32.8 | 43.4 | 0.0 | 13.3 | 10.0 | 67.4 | 29.8 | 37.5 |
| Qwen2.5-Math-7B-Instruct (Yang et al., 2024b) | 7B | 95.2 | 73.8 | 38.0 | 22.0 | 13.0 | 10.0 | 13.3 | 54.8 | 28.3 | 38.7 |
| Llama-3.1-8B-Instruct (Grattafiori et al., 2024) | 8B | 84.5 | 52.2 | 15.0 | 10.0 | 3.0 | 2.0 | 3.0 | 68.0 | 30.0 | 29.7 |
| DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI, 2025) | 8B | 83.0 | 89.1 | 48.0 | 25.0 | 50.4 | 35.0 | 25.0 | 52.0 | 49.0 | 50.7 |
| Qwen2.5-7B (Base) (Yang et al., 2024a) | 7B | 85.4 | 49.8 | 25.0 | 18.0 | 7.0 | 5.0 | 8.0 | 74.2 | 36.4 | 34.3 |
| + SFT | 7B | 91.3 | 65.4 | 42.5 | 54.2 | 16.7 | 12.2 | 16.7 | 70.1 | 28.3 | 44.2 |
| + SimpleRL-Zoo (Zeng et al., 2023) | 7B | 91.7 | 78.2 | 38.6 | 40.4 | **20.0** | 15.6 | 15.0 | 62.5 | **55.2** | 46.4 |
| + Open-Reasoner-Zero (Hu et al., 2026) | 7B | 89.0 | **81.4** | 47.9 | 38.4 | 17.9 | 15.6 | 13.3 | 62.5 | 36.6 | 44.6 |
| + TopoPRM (Ours) | 7B | **95.8** | 72.6 | **47.8** | **57.2** | 16.7 | **26.7** | **47.0** | **82.7** | 51.0 | **55.3** |
| DeepSeek-R1-Distill-Qwen-7B (Base) (DeepSeek-AI, 2025) | 7B | 73.8 | 36.8 | 25.8 | 28.0 | 46.7 | 30.0 | 22.9 | 23.7 | 12.6 | 33.4 |
| + SFT | 7B | **86.0** | **92.8** | 52.0 | 48.0 | 55.5 | 40.0 | 30.0 | 50.0 | **49.1** | 55.9 |
| + GRPO (Shao et al., 2024) | 7B | 85.1 | **67.4** | 56.2 | 72.0 | 50.0 | 40.0 | **57.8** | 50.1 | 17.7 | 55.1 |
| + TopoPRM (Ours) | 7B | 84.3 | 66.6 | **56.4** | **72.2** | **55.5** | **40.0** | 55.4 | **60.3** | 34.3 | **58.3** |
| Qwen3.5-9B (Base) | 9B | 69.5 | 66.6 | 55.2 | 61.5 | 16.7 | 21.1 | 16.7 | 28.6 | 37.4 | 41.5 |
| + SFT | 9B | 94.1 | 68.2 | 44.0 | 53.8 | 30.0 | 15.6 | 30.0 | 78.4 | **47.0** | 51.2 |
| + GRPO (Shao et al., 2024) | 9B | 93.3 | 63.0 | 56.2 | **72.0** | **46.7** | **30.0** | **57.8** | 63.0 | 17.7 | 55.5 |
| + TopoPRM (Ours) | 9B | **97.9** | **81.9** | **56.2** | 76.2 | 43.3 | 18.9 | 47.0 | **86.4** | 40.4 | **60.9** |

Table 14: Archived EMNLP Table 9: reward aggregation comparison on the in-domain critique set. All numeric values are retained from the submission.

| **Strategy** | **Mid** | **High** | **Fmt%** | **Len** | **Collapse%↓** |
|---|---|---|---|---|---|
| Linear (weighted sum) | 37.2 | 22.3 | 94.6 | 364 | 75.1 |
| Clipped linear | 29.0 | 21.5 | 93.0 | 382 | 67.7 |
| Legacy hierarchical aggregation | 37.2 | 22.3 | 94.6 | 364 | 37.9 |

Table 12: Archived EMNLP Table 3: reward component ablation on Qwen3.5-9B. Avg is the originally submitted unweighted mean pass@1 across eight benchmarks.

| Reward Config. | Avg (%) | $\Delta$ | Tok | Acc/kTok↑ | Collapse%↓ |
|---|---|---|---|---|---|
| TopoPRM (Full) | 45.6 | — | 1692 | 26.9 | 37.9 |
| w/o ACE | 41.7 | −3.8 | 1958 | 21.3 | 42.1 |
| w/o Topology | 41.8 | −3.8 | 1943 | 21.5 | 44.6 |
| w/o Continuity | 16.3 | −29.3 | 2047 | 8.0 | 68.8 |
| Outcome Only | 40.8 | −4.7 | 1823 | 22.4 | 57.8 |

Table 13: Archived EMNLP Table 4: cross-scale distillation from the 9B TopoPRM teacher. Token Ratio is mean generation length relative to the teacher; Structural Retention aggregates the originally submitted DAG-profile metrics on MATH-500.

| Scale | GSM8K | MATH | Tok | Ratio | Struct. Ret. |
|---|---|---|---|---|---|
| 9B (Teacher) | 84.3 | 66.6 | 679 | 1.00× | 1.00× |
| 4B | 82.8 | 61.5 | 382 | 0.56× | 0.93× |
| 2B | 78.9 | 57.1 | 305 | 0.45× | 0.88× |
| 0.8B | 73.6 | 52.4 | 244 | 0.36× | 0.81× |
| 4B (SFT-distill) | 79.4 | 58.2 | 438 | 0.64× | 0.86× |
| 4B (Off-policy KL) | 76.8 | 49.3 | 4096 | >1.0× | 0.41× |

Table 15: Per-benchmark results and component ablations of TopoPRM. We report **p@1**, **p@5**, **maj@5**, mean generated tokens / total wall-clock seconds (**Tok / Time**, in ks), and token-efficiency **Acc/kTok** $= \text{p@1}/(\overline{\text{tokens}}/1000)$, We use the same protocol in A. **Bold** / underline: best / second-best per column.

| Dataset / Method | GSM8K (1,319) | | | | | MATH-500 (500) | | | | | OlympiadBench (134) | | | | | Omni-MATH (262) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | p@1 | p@5 | maj@5 | Tok / Time | Acc/ kTok | p@1 | p@5 | maj@5 | Tok / Time | Acc/ kTok | p@1 | p@5 | maj@5 | Tok / Time | Acc/ kTok | p@1 | p@5 | maj@5 | Tok / Time | Acc/ kTok |
| *Base (Qwen3.5-9B)* | 55.2 | 87.8 | 55.7 | 1536/45.0 | 35.9 | 19.8 | 39.2 | 21.4 | 1536/18.7 | 12.9 | 11.2 | 25.4 | 14.2 | 2560/26.5 | 4.4 | 16.4 | 33.2 | 17.9 | 2559/52.0 | 6.4 |
| Outcome Only | 93.3 | 97.7 | 93.6 | 1047/34.2 | 89.1 | 50.8 | 59.0 | 55.2 | 1511/18.9 | 33.6 | 31.3 | 46.3 | 37.3 | 2043/19.7 | 15.3 | 41.6 | 55.3 | **48.9** | 1845/35.2 | 22.5 |
| w/o Topology | 93.0 | 97.7 | 93.9 | 946/25.0 | 98.3 | 51.0 | 59.4 | **55.6** | 1506/16.2 | 33.9 | 29.8 | 50.7 | 38.1 | 2468/11.8 | 12.1 | 42.4 | 54.2 | 45.8 | 2430/17.8 | 17.4 |
| w/o Continuity | 51.0 | 86.1 | 52.6 | 1528/48.3 | 33.4 | 21.6 | 39.6 | 23.2 | 1536/19.1 | 14.1 | 8.2 | 25.4 | 9.0 | 2560/ 8.7 | 3.2 | 14.9 | 30.5 | 17.6 | 2560/18.5 | 5.8 |
| w/o ACE | 93.8 | **98.0** | 94.6 | 1006/28.3 | 93.2 | 50.8 | 59.8 | 55.2 | 1515/18.2 | 33.5 | 32.8 | 47.0 | 35.8 | 2491/ 9.6 | 13.2 | 42.0 | 53.8 | 47.3 | 2463/19.4 | 17.1 |
| **★ TopoPRM (Full)** | **94.1** | 97.9 | **94.8** | **796/19.7** | **118.2** | **51.0** | **81.9** | 54.6 | **1454/16.6** | **35.1** | **32.8** | **50.7** | **38.8** | **1768/18.1** | **18.6** | **42.8** | **55.7** | 47.7 | **1592/19.0** | **26.9** |

| Dataset / Method | AIME 2024 (30) | | | | | AIME 2025 (90) | | | | | CNMO 2024 (30) | | | | | MMLU (500) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | p@1 | p@5 | maj@5 | Tok / Time | Acc/ kTok | p@1 | p@5 | maj@5 | Tok / Time | Acc/ kTok | p@1 | p@5 | maj@5 | Tok / Time | Acc/ kTok | p@1 | p@5 | maj@5 | Tok / Time | Acc/ kTok |
| *Base (Qwen3.5-9B)* | 16.7 | 33.3 | 33.3 | 4083/ 8.5 | 4.1 | 2.2 | 4.4 | 1.1 | 2560/ 5.9 | 0.9 | 10.0 | 16.7 | 3.3 | 2560/ 1.9 | 3.9 | 28.6 | 41.4 | 26.2 | 512/55.1 | 55.9 |
| Outcome Only | 16.7 | 33.3 | 20.0 | 2546/ 5.6 | 6.6 | 13.3 | 15.6 | 8.9 | 2547/16.7 | 5.2 | 16.7 | 33.3 | 26.7 | 2530/ 5.5 | 6.6 | 63.0 | 84.0 | 61.0 | 512/146.2 | 123.0 |
| w/o Topology | 20.0 | 40.0 | 30.0 | 2560/ 2.1 | 7.8 | 12.2 | 16.7 | 10.0 | 2560/ 6.2 | 4.8 | 20.0 | **40.0** | 26.7 | 2560/ 2.0 | 7.8 | 65.8 | 85.0 | 63.0 | 512/32.3 | 128.5 |
| w/o Continuity | 6.7 | 13.3 | 3.3 | 2560/ 2.3 | 2.6 | 1.1 | 4.4 | 0.0 | 2560/ 7.0 | 0.4 | 6.7 | 20.0 | 3.3 | 2560/ 2.4 | 2.6 | 20.2 | 34.8 | 18.6 | 512/80.4 | 39.5 |
| w/o ACE | 20.0 | 30.0 | 23.3 | 2560/ 2.6 | 7.8 | 13.3 | 17.8 | **12.2** | 2560/ 7.7 | 5.2 | 20.0 | 30.0 | 26.7 | 2560/ 2.6 | 7.8 | 61.1 | 83.1 | 61.2 | 508/58.1 | 120.3 |
| **★ TopoPRM (Full)** | **30.0** | **43.3** | **33.3** | **2431/ 5.4** | **12.3** | **15.6** | **18.9** | 11.1 | **2457/ 7.7** | **6.4** | **30.0** | **47.0** | **30.0** | **2530/ 2.5** | **11.9** | **68.2** | **86.4** | **68.8** | **508/48.0** | **134.2** |

Table 16: Main results on the in-domain mathematical critique benchmark. **Middle** and **High** denote middle-school and high-school subsets. Acc measures critique scoring accuracy; Fmt% measures structural compliance; Len reports mean prediction length in tokens. Best results per column are in **bold**, second-best are <u>underlined</u>.

| | Middle School | | | High School | | | |
|---|---|---|---|---|---|---|---|
| **Model** | Acc | Fmt% | Len | Acc | Fmt% | Len | **Overall** |
| *Zero-shot Baseline* | | | | | | | |
| Qwen3-32B | 16.0 | 64.5 | 1 200 | 7.0 | 43.5 | 1 240 | 11.5 |
| Qwen3-32B + SFT | 29.0 | 78.5 | 1 164 | 19.5 | 65.0 | 1 233 | 24.3 |
| *RFT Variants* | | | | | | | |
| + Outcome Only | 19.5 | 89.0 | 410 | 13.0 | 85.5 | 475 | 16.3 |
| + w/o Topology | 23.0 | 91.0 | 397 | 16.5 | 86.0 | 460 | 19.8 |
| + w/o Continuity | 27.5 | 90.5 | 379 | 20.0 | 85.0 | 436 | 23.8 |
| + Clipped Linear | 29.0 | 93.0 | 382 | 21.5 | 88.5 | 458 | 25.3 |
| + TopoPRM (full) | **37.2** | **94.6** | **364** | **22.3** | **89.3** | **424** | **29.7** |
| *Topology-Guided Distillation Fine-tuned* | | | | | | | |
| Qwen3.5-9B + full | <u>40.5</u> | **98.0** | 1 602 | **30.5** | **94.0** | 2 191 | **35.5** |
| Qwen3-8B + full | **60.0** | <u>94.0</u> | 532 | **58.0** | <u>97.0</u> | 613 | **59.0** |

Table 17: Reward component ablation on **Qwen3-32B** over the in-domain critique validation set. Each configuration toggles the topology ($r_{\text{topo}}$) and continuity ($r_{\text{cont}}$) signals on top of the outcome reward $r_{\text{out}}$. **Avg** is the unweighted mean of Middle- and High-school accuracy; $\Delta$ denotes the gap relative to the full TopoPRM. Rows are sorted by Avg in ascending order to highlight the monotonic contribution of each component.

| | **Reward Components** | | | **Accuracy (%)** | | | |
|---|---|---|---|---|---|---|---|
| **Configuration** | $r_{\text{out}}$ | $r_{\text{topo}}$ | $r_{\text{cont}}$ | Mid | High | **Avg** | $\Delta$ |
| Outcome Only | ✓ | ✗ | ✗ | 19.5 | 13.0 | 16.3 | −13.0 |
| w/o Topology | ✓ | ✗ | ✓ | 23.0 | 16.5 | 19.8 | − 9.5 |
| w/o Continuity | ✓ | ✓ | ✗ | 27.5 | 20.0 | 23.8 | − 5.5 |
| **TopoPRM (Full)** | ✓ | ✓ | ✓ | **33.0** | **25.5** | **29.3** | — |

Table 18: Accuracy-length trade-off on the in-domain critique validation set (Qwen3-32B). Acc/k-Tok = Overall-Acc per 1,000 tokens. Len↑ is the length increase relative to TopoPRM (lower is better).

| | **Middle** | | **High** | | **Efficiency** | |
|---|---|---|---|---|---|---|
| **Config.** | Acc | Len | Acc | Len | Acc/kTok↑ | Len↑ |
| TopoPRM (full) | 37.2 | 364 | 22.3 | 424 | 75.5 | — |
| w/o Continuity | 27.5 | 379 | 20.0 | 436 | 58.4 | +4.1% |
| w/o Topology | 23.0 | 397 | 16.5 | 460 | 46.2 | +9.0% |
| Outcome Only | 19.5 | 410 | 13.0 | 475 | 36.8 | +12.6% |
| Qwen3-32B (base) | 16.0 | 1200 | 7.0 | 1240 | 9.4 | +230% |

Table 19: Structural quality of extracted reasoning DAGs on the in-domain middle-school evaluation set. Acyclic% is the fraction of valid DAGs; No-Orphan% is the fraction of traces without unsupported conclusion nodes; Dir-Cons. is mean directional consistency; DAG Depth is mean dependency depth; Edge-Keep% is the fraction of dependency edges retained after compression.

| **Model** | **Acyclic%↑** | **No-Orphan%↑** | **Dir-Cons.↑** | **DAG Depth** | **Edge-Keep%↑** |
|---|---|---|---|---|---|
| Qwen3-32B + SFT | 100 | 61 | 0.49 | 1.05 | 43 |
| + GRPO (outcome only) | 100 | 82 | 0.89 | 1.12 | 56 |
| + GRPO (w/o topology) | 100 | 84 | 0.91 | 1.14 | 58 |
| + GRPO (w/o continuity) | 100 | 83 | 0.91 | 1.13 | 57 |
| + GRPO (TopoPRM full) | 100 | 89 | 1.00 | 1.17 | 60 |

### H.1 Additional EMNLP-workspace results

The following six supplementary tables were present in the accompanying EMNLP source workspace but were not among the nine result tables rendered in the submission PDF. Their values are retained for reference under the same archival restrictions.

Table 20: Accuracy evaluation across 9 public math reasoning benchmarks. All our models follow one `transformers`-backend protocol with the model's native chat template, shared answer extraction, and shared matching. Open-reported numbers are taken from official papers and model cards. **Bold** marks the best result per block and underline marks the second best. This is the retained EMNLP-submission table: entries prefixed by $\sim$ are archival approximations, not ICLR claims, and all rows are being revalidated under the canonical protocol.

| Model | Params | Elem. Math | | Olympiad Math | | Competition | | CNMO | General | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | GSM8K | MATH-500 | Olympiad | Omni-MATH | AIME'24 | AIME'25 | CNMO'24 | MMLU | GPQA-D |
| *Reference open-source models (open-reported)* | | | | | | | | | | |
| Qwen2.5-7B-Instruct (Yang et al., 2024a) | 7B | 91.6 | 75.5 | $\sim$25.0 | $\sim$18.0 | $\sim$7.0 | $\sim$5.0 | $\sim$8.0 | 75.4 | 36.4 |
| Qwen2.5-Math-7B-Instruct (Yang et al., 2024b) | 7B | 95.2 | 83.6 | $\sim$38.0 | $\sim$22.0 | $\sim$13.0 | $\sim$10.0 | $\sim$15.0 | $\sim$60.0 | $\sim$35.0 |
| Llama-3.1-8B-Instruct (Grattafiori et al., 2024) | 8B | 84.5 | 52.2 | $\sim$15.0 | $\sim$10.0 | $\sim$3.0 | $\sim$2.0 | $\sim$3.0 | $\sim$68.0 | $\sim$30.0 |
| DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI, 2025) | 8B | $\sim$83.0 | 89.1 | $\sim$48.0 | $\sim$25.0 | 50.4 | $\sim$35.0 | $\sim$25.0 | $\sim$52.0 | 49.0 |
| DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) | 7B | $\sim$86.0 | 92.8 | $\sim$52.0 | $\sim$28.0 | 55.5 | $\sim$40.0 | $\sim$30.0 | $\sim$55.0 | 49.1 |
| SimpleRL-Zoo (Qwen2.5-7B) (Zeng et al., 2023) | 7B | 91.7 | 78.2 | $\sim$38.6 | $\sim$40.4 | 20.0 | $\sim$15.6 | $\sim$15.0 | $\sim$62.5 | $\sim$55.2 |
| *Ours: 9B family (Qwen3.5-9B base, unified transformers protocol)* | | | | | | | | | | |
| Qwen3.5-9B (base) | 9B | 67.1 | 43.8 | 29.8 | 51.4 | 6.7 | 3.3 | 12.1 | 52.9 | 27.3 |
| + SFT | 9B | 94.1 | 50.8 | 32.8 | 42.4 | 30.0 | 15.6 | 30.0 | 68.2 | 20.7 |
| + GRPO (outcome-only) | 9B | 93.3 | 50.8 | 31.3 | 41.6 | 16.7 | 13.3 | 16.7 | 63.0 | – |
| + GRPO (w/o topology) | 9B | 93.0 | 51.0 | 29.8 | 42.4 | 20.0 | 12.2 | 20.0 | 65.8 | 16.2 |
| + GRPO (w/o continuity) | 9B | – | – | – | – | – | – | – | – | – |
| + TopoPRM (hierarchical) | 9B | **93.5** | 49.8 | **32.8** | **42.8** | **26.7** | 12.2 | **26.7** | 61.8 | 16.7 |
| + TopoPRM (gated) | 9B | 93.8 | **50.8** | **32.8** | 42.0 | 20.0 | **13.3** | 20.0 | 61.1 | **21.2** |
| *Ours: DR1-7B family (DeepSeek-R1-Distill-Qwen-7B base, unified transformers protocol)* | | | | | | | | | | |
| DR1-7B (base, measured) | 7B | 60.8 | 68.4 | 57.8 | **72.8** | 46.7 | 33.3 | 23.3 | 42.4 | 16.2 |
| + SFT | 7B | 73.8 | 36.8 | 25.8 | 48.0 | 0.0 | 0.0 | 22.9 | 23.7 | 12.6 |
| + GRPO (outcome-only) | 7B | **85.1** | 67.4 | 56.2 | 72.0 | 46.7 | 30.0 | 57.8 | 50.1 | **17.7** |
| + GRPO (w/o topology) | 7B | 84.5 | **68.8** | – | – | 36.7 | 33.3 | 53.0 | – | 11.1 |
| + GRPO (w/o continuity) | 7B | **85.1** | 66.4 | – | – | 36.7 | 26.7 | 50.6 | – | 14.6 |
| + TopoPRM (hierarchical) | 7B | 84.3 | 66.6 | 56.4 | 72.2 | **50.0** | **40.0** | 55.4 | 49.6 | 15.2 |
| *Ours: Distilled students (Topology-Guided Distillation, target $\leq$4B)* | | | | | | | | | | |
| Student (4B, SFT distill) | 4B | $\sim$89.4 | $\sim$49.8 | $\sim$24.0 | $\sim$30.2 | $\sim$12.1 | $\sim$8.5 | $\sim$6.8 | $\sim$82.6 | $\sim$40.7 |
| Student (4B, Topology-Guided Distillation) | 4B | $\sim$91.3 | $\sim$52.1 | $\sim$27.6 | $\sim$34.8 | $\sim$15.4 | $\sim$10.2 | $\sim$8.1 | $\sim$83.5 | $\sim$41.9 |
| Student (2B, Topology-Guided Distillation) | 2B | $\sim$86.7 | $\sim$46.5 | $\sim$21.3 | $\sim$27.1 | $\sim$9.2 | $\sim$6.4 | $\sim$5.7 | $\sim$80.8 | $\sim$38.4 |
| Student (0.8B, Topology-Guided Distillation) | 0.8B | $\sim$79.8 | $\sim$39.4 | $\sim$14.0 | $\sim$18.6 | $\sim$6.1 | $\sim$4.5 | $\sim$3.9 | $\sim$76.3 | $\sim$34.2 |

Table 21: Detailed unified metrics across our measured configurations (`num_samples=5`, native chat template, same extractor). **Cor** = correct count; **Err** = error count; **Tok** = mean generation tokens; **Acc/kTok** = pass@1 points per 1 000 tokens (higher is more token-efficient). Best within each block in **bold**.

| | | **GSM8K** | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Model** | **Params** | pass@1 | pass@5 | maj@5 | prm@5 | F1 | Cor/Err | Tok | Acc/kTok |
| *DR1-7B family* | | | | | | | | | |
| DR1-7B (base, measured) | 7B | 60.8 | 80.8 | 64.2 | 60.8 | 60.8 | 800/519 | 617 | 98.5 |
| + SFT | 7B | 73.8 | 91.5 | 80.2 | 73.7 | 73.8 | 970/349 | 634 | 116.4 |
| + GRPO (outcome-only) | 7B | **85.1** | 94.0 | **88.3** | **85.1** | 85.1 | 1120/199 | 674 | 126.3 |
| + GRPO (w/o topology) | 7B | 84.5 | 93.9 | 87.5 | 84.5 | 84.5 | 1113/206 | 686 | 123.2 |
| + GRPO (w/o continuity) | 7B | **85.1** | **94.2** | 87.9 | 85.1 | 85.1 | 1121/198 | 640 | 133.0 |
| + TopoPRM (hierarchical) | 7B | 84.3 | 93.6 | 88.3 | 84.3 | 84.3 | 1110/209 | 679 | 124.2 |
| *9B family (our unified protocol)* | | | | | | | | | |
| Qwen3.5-9B (base) | 9B | 67.1 | – | – | – | 67.1 | 850/469 | 1536 | 43.7 |

| | | **MATH-500** | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Model** | **Params** | pass@1 | pass@5 | maj@5 | prm@5 | F1 | Cor/Err | Tok | Acc/kTok |
| *DR1-7B family* | | | | | | | | | |
| DR1-7B (base, measured) | 7B | **68.4** | 70.6 | 67.4 | **68.4** | 68.4 | 322/178 | 4957 | 13.8 |
| + SFT | 7B | 36.8 | 54.8 | 43.4 | 36.8 | 36.8 | 185/315 | 3731 | 9.9 |
| + GRPO (outcome-only) | 7B | 67.4 | 72.6 | 67.4 | 67.4 | 67.4 | 318/182 | 6001 | 11.2 |
| + GRPO (w/o topology) | 7B | **68.8** | **73.2** | **68.8** | **68.8** | 68.8 | 324/176 | 5973 | 11.5 |
| + GRPO (w/o continuity) | 7B | 66.4 | 71.8 | 67.8 | 66.4 | 66.4 | 312/188 | 6088 | 10.9 |
| + TopoPRM (hierarchical) | 7B | 66.6 | 72.8 | 68.2 | 66.6 | 66.6 | 314/186 | 6053 | 11.0 |
| *9B family (our unified protocol)* | | | | | | | | | |
| Qwen3.5-9B (base) | 9B | 43.8 | – | – | – | 43.8 | 203/297 | 3072 | 14.3 |

| | | **AIME 2024** | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Model** | **Params** | pass@1 | pass@5 | maj@5 | prm@5 | F1 | Cor/Err | Tok | Acc/kTok |
| *DR1-7B family* | | | | | | | | | |
| DR1-7B (base, measured) | 7B | 46.7 | **46.7** | 36.7 | 46.7 | 46.7 | 10/20 | 8138 | 5.7 |
| + SFT | 7B | 0.0 | 3.3 | 0.0 | 0.0 | 0.0 | 0/30 | 4301 | 0.0 |
| + GRPO (outcome-only) | 7B | 46.7 | 36.7 | 33.3 | 46.7 | 46.7 | 9/21 | 8192 | 5.7 |
| + GRPO (w/o topology) | 7B | 36.7 | **46.7** | 33.3 | 36.7 | 36.7 | 8/22 | 8192 | 4.5 |
| + GRPO (w/o continuity) | 7B | 36.7 | 40.0 | 33.3 | 36.7 | 36.7 | 8/22 | 8192 | 4.5 |
| + TopoPRM (hierarchical) | 7B | **50.0** | 40.0 | 33.3 | **50.0** | 50.0 | 10/20 | 8192 | **6.1** |
| *9B family* | | | | | | | | | |
| Qwen3.5-9B (base) | 9B | 6.7 | – | – | – | 6.7 | 1/29 | 4096 | 1.6 |

| | | **Omni-MATH (500 subset)** | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Model** | **Params** | pass@1 | pass@5 | maj@5 | prm@5 | F1 | Cor/Err | Tok | Acc/kTok |
| *DR1-7B family* | | | | | | | | | |
| DR1-7B (base, measured) | 7B | **72.8** | – | – | **72.8** | 72.8 | 348/152 | 4720 | 15.4 |
| + SFT | 7B | 48.0 | – | – | 48.0 | 48.0 | 235/265 | 7114 | 6.7 |
| + GRPO (outcome-only) | 7B | 72.0 | – | – | 72.0 | 72.0 | 347/153 | 3562 | 20.2 |
| + TopoPRM (hierarchical) | 7B | 72.2 | – | – | 72.2 | 72.2 | 348/152 | 3529 | **20.5** |

Table 22: Ablation of TopoPRM reward components on the Qwen3.5-9B backbone. All variants start from the same SFT checkpoint and run identical GRPO steps. We report pass@1 / pass@5 (%), mean generation tokens (Tok), and Acc/kTok = pass@1 per 1 000 tokens. $\Delta$ shows the change in pass@5 relative to the full TopoPRM (hierarchical). Best in **bold**.

| | **GSM8K** | | | | **MATH-500** | | | | **AIME'24** | | | | **Omni-MATH** | | | | **MMLU** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Reward Config** | p@1 | p@5 | Tok | $\Delta$p@5 | p@1 | p@5 | Tok | $\Delta$p@5 | p@1 | p@5 | Tok | $\Delta$p@5 | p@1 | p@5 | Tok | $\Delta$p@5 | p@1 | p@5 | Tok | $\Delta$p@5 |
| TopoPRM (full, hier.) | 93.5 | **97.9** | 1026 | — | 49.8 | 59.0 | 1506 | — | **26.7** | **43.3** | 2560 | — | **42.8** | **55.7** | 2465 | — | 61.8 | 81.9 | 508 | — |
| TopoPRM (gated) | 93.8 | 97.9 | 1006 | +0.1 | **50.8** | **59.8** | 1515 | +0.8 | 20.0 | 30.0 | 2560 | −13.3 | 42.0 | 53.8 | 2463 | −1.9 | 61.1 | **83.1** | 508 | +1.3 |
| w/o Topology | 93.0 | 97.7 | 946 | −0.2 | 51.0 | 59.4 | 1506 | +0.4 | 20.0 | 40.0 | 2560 | −3.3 | 42.4 | 54.2 | 2430 | −1.5 | **65.8** | 85.0 | 512 | +3.1 |
| w/o Continuity | 51.0 | 86.1 | 1528 | −11.8 | 21.6 | 39.6 | 1536 | −19.4 | 6.7 | 13.3 | 2560 | −30.0 | 14.9 | 30.5 | 2560 | −25.2 | 20.2 | 34.8 | 512 | −47.1 |
| Outcome Only GRPO | 93.3 | 97.7 | 1047 | −0.2 | 50.8 | 59.0 | 1511 | +0.0 | 16.7 | 33.3 | 2546 | −10.0 | 41.6 | 55.3 | 1845 | −0.4 | 63.0 | 84.0 | 512 | +2.1 |
| SFT baseline | **94.1** | 97.7 | **796** | −0.2 | 50.8 | 60.4 | **1454** | +1.4 | 30.0 | **43.3** | 2431 | +0.0 | 42.4 | 53.1 | **1592** | −2.7 | 68.2 | **86.4** | 512 | +4.5 |

Table 23: Generation efficiency of Qwen3.5-9B variants. All runs use 5 samples per item on the same hardware. **Tok** = mean generation tokens per sample; **Time** = total wall-clock seconds for the benchmark; **Tok/s** = throughput (total tokens / time); **Acc/kTok** = pass@1 per 1 000 tokens (higher = more token-efficient). Best in **bold**.

| | GSM8K (1,319 items) | | | | MATH-500 | | | | AIME'24 (30 items) | | | | Omni-MATH (262 items) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Model** | Tok | Time | Tok/s | Acc/kTok | Tok | Time | Tok/s | Acc/kTok | Tok | Time | Tok/s | Acc/kTok | Tok | Time | Tok/s | Acc/kTok |
| **TopoPRM (Full)** | **796** | **19.7k** | **266** | **118.2** | **1454** | **16.6k** | **220** | **34.9** | **2431** | 5.4k | 67 | 12.3 | **1592** | **31.7k** | **66** | **26.6** |
| Outcome Only GRPO | 1047 | 34.2k | 202 | 89.1 | 1511 | 18.9k | 200 | 33.6 | 2546 | 5.6k | 68 | 6.5 | 1845 | 35.2k | 69 | 22.5 |
| w/o Topology | 946 | 25.0k | 250 | 98.3 | 1506 | 16.2k | 232 | 33.9 | 2560 | **2.1k** | **186** | 7.8 | 2430 | 17.8k | **179** | 17.4 |
| w/o Continuity | 1528 | 48.3k | 209 | 33.4 | 1536 | 19.1k | 201 | 14.1 | 2560 | 2.3k | 165 | 2.6 | 2560 | 18.5k | 181 | 5.8 |
| Legacy gated agg. | 1006 | 28.3k | 234 | 93.2 | 1515 | 18.2k | 208 | 33.5 | 2560 | 2.6k | 147 | 7.8 | 2463 | 19.4k | 167 | 17.0 |
| TopoPRM (Hier.) | 1026 | 33.4k | 203 | 91.1 | 1506 | 18.0k | 209 | 33.1 | 2560 | 2.6k | 149 | **10.4** | 2465 | 19.0k | 170 | 17.3 |

Table 24: First-Attempt vs Revised accuracy on 1K sampled AIME-2024 / MATH problems (Qwen3.5-9B family). First Attempt: one sample from the model. Revised Attempt: a second sample conditioned on the first attempt and a topology-aware revision prompt $P_r$ (Eq. 6). Baseline = plain distillation reviser without DAG signal. Avg-Len refers to the revised response only.

| Model | First Att. | Revised Att. | Gain | Revised Len | Revised / First |
|---|---|---|---|---|---|
| Qwen3.5-9B (base, no SRT) | $\sim$31.8 | $\sim$34.6 | $\sim$+2.8 | $\sim$742 | $\sim$1.07 |
| + Plain distillation reviser (no topology) | $\sim$33.1 | $\sim$37.0 | $\sim$+3.9 | $\sim$688 | $\sim$1.02 |
| + Topology-Guided Distillation reviser (Eq. 6) | $\sim$33.0 | $\sim$39.8 | $\sim$+6.8 | $\sim$624 | $\sim$0.94 |

Table 25: Case study comparing teacher and topology-guided distilled-student traces on the same prompt.

| | **Teacher (TopoPRM)** | **Distilled student** |
|---|---|---|
| Problem | High-school algebra problem requiring multi-step substitution and verification (same prompt hash for both rows). | |
| # Steps | $\sim$14 | $\sim$9 |
| # Tokens | $\sim$612 | $\sim$428 |
| Score-Acc | $\sim$0.93 | $\sim$0.91 |
| Key reasoning | Core conclusion supported by an explicit minimal chain of derivations and substitutions. | Preserves key support edges while compressing redundant branches and intermediate restatements. |